

20. Data Transfer Controller (DTCb)

This MCU incorporates a data transfer controller (DTC).

The DTC is triggered by an interrupt request to perform data transfers.

In addition to the conventional methods of DTC transfer (normal, repeat, block, and chain), DTCb supports sequential transfer, in which it handles a series of transfers made up of a combination of the other methods. In sequential transfer, the data that is initially transferred selects one from possible 256 sequences for execution. The DTCb can divide one sequence into several transfers depending on how the parts of the sequence are combined.

20.1 Overview

Table 20.1 lists the specifications of the DTC, and Figure 20.1 shows a block diagram of the DTC.

Table 20.1 DTC Specifications

Item	Description
Number of transfer channels	The same number as all interrupt sources that can start the DTC transfer.
Transfer modes	- Normal transfer mode A single transfer request leads to a single data transfer. - Repeat transfer mode A single transfer request leads to a single data transfer. The transfer address is returned to the transfer start address after the number of data transfers corresponding to "repeat size". The maximum number of repeat transfers is 256, and the maximum data transfer size is 256×32 bits, 1024 bytes. - Block transfer mode A single transfer request leads to the transfer of a single block. The maximum block size is 256×32 bits = 1024 bytes.
Chain transfer	- Multiple types of data transfers can sequentially be executed in response to a single request. - Either "performed only when the transfer counter becomes 0" or "every time" can be selected.
Sequence transfer	A series of complicated transfers can be registered as a sequence. Any sequence can be selected by the transfer data and executed. - Only one trigger source can be set at a time. - Up to 256 sequences for a single trigger source - The data that is initially transferred in response to a transfer request determines a sequence - The whole sequence can be executed on a single request, or be suspended in the middle of the sequence and resumed on the next transfer request (division of sequence).
Transfer space	- In short-address mode: 16 Mbytes (Areas from 0000 0000h to 007F FFFFh and FF80 0000h to FFFF FFFFh except reserved areas) - In full-address mode: 4 Gbytes (Area from 0000 0000h to FFFF FFFFh except reserved areas)
Data transfer units	- Single data: 1 byte (8 bits), 1 word (16 bits), 1 longword (32 bits) - Single block size: 1 to 256 data
CPU interrupt source	- An interrupt request can be generated to the CPU on a request source for a data transfer. - An interrupt request can be generated to the CPU after a single data transfer. - An interrupt request can be generated to the CPU after data transfer of specified volume.
Event link function	An event link request is generated after one data transfer (for block, after one block transfer).
Read skip	Reading of the transfer information can be skipped when the same transfer is repeated.
Write-back skip	Write-back of the transferred data that is not updated can be skipped when the address of the transfer source or destination is fixed.
Write-back disable	Allows disabling the write-back of transfer information.
Displacement addition	The displacement value can be added to the transfer source address (for each transfer information)
Low power consumption function	Module stop state can be set.

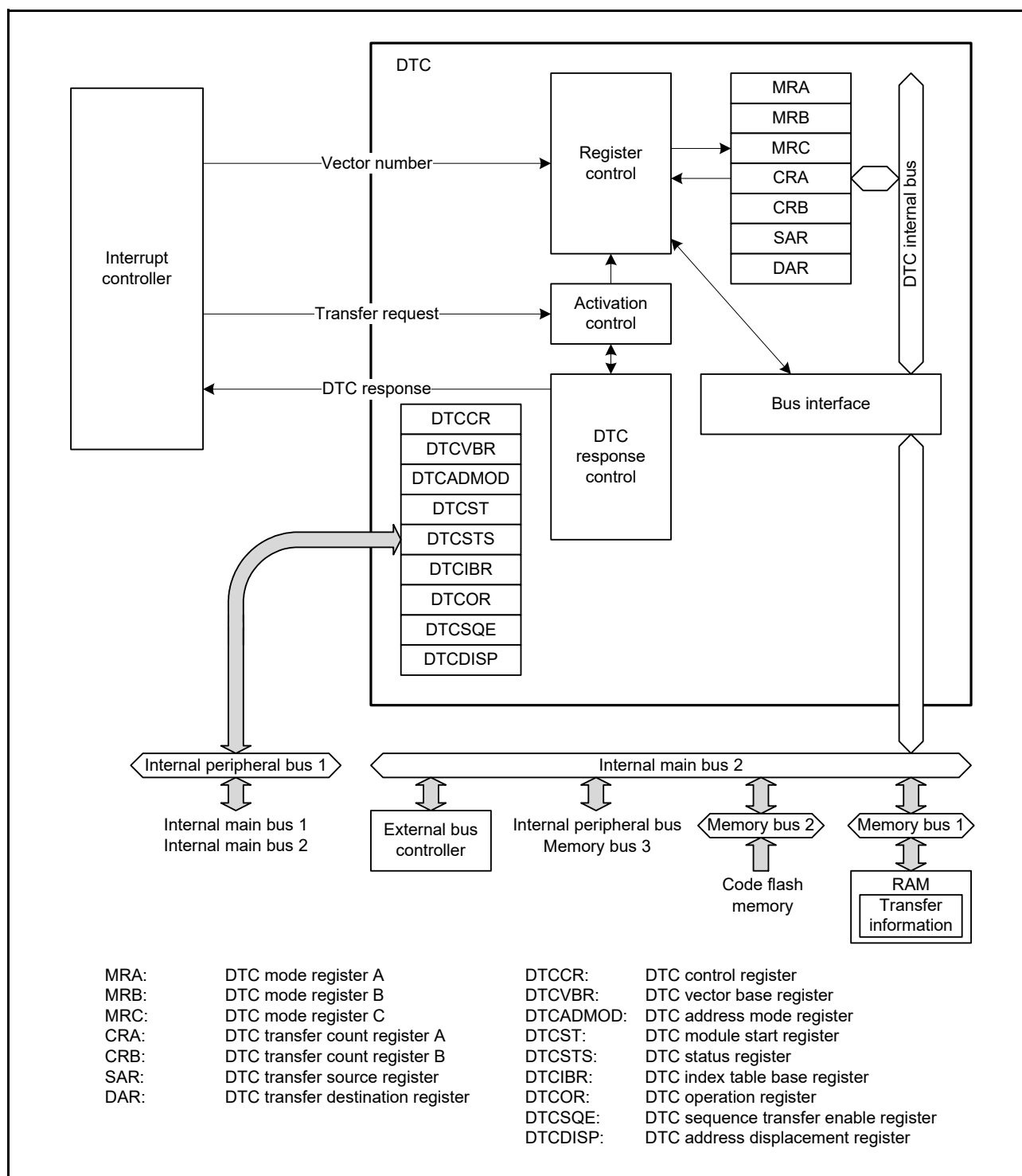


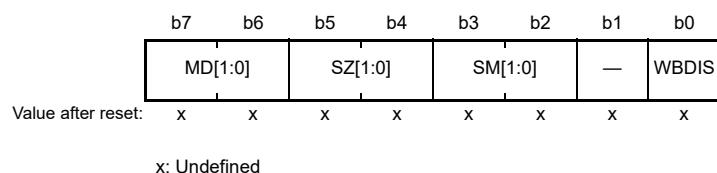
Figure 20.1 DTC Block Diagram

20.2 Register Descriptions

Registers MRA, MRB, MRC, SAR, DAR, CRA, and CRB are DTC internal registers, which cannot be directly accessed from the CPU. Values to be set in these DTC internal registers are placed in the RAM area as transfer information. When accepting a transfer request, the DTC reads the transfer information from the RAM area and sets it in the internal registers. After the data transfer ends, the values of the updated internal register are written back to the RAM area as transfer information.

20.2.1 DTC Mode Register A (MRA)

Address(es): (inaccessible directly from the CPU)



Bit	Symbol	Bit Name	Description	R/W
b0	WBDIS	Write-back Disable	0: Writes back the transfer information on completion of the data transfer 1: Does not write back the transfer information on completion of the data transfer	—
b1	—	Reserved	Set this bit to 0.	—
b3, b2	SM[1:0]	Transfer Source Address Addressing Mode	b3 b2 0 0: The address in the SAR register is fixed. (write-back to SAR is skipped.) 0 1: The address in the SAR register is fixed. (write-back to SAR is skipped.) 1 0: The SAR value is incremented after a data transfer. (+1 when the SZ[1:0] bits are 00b, +2 when 01b, +4 when 10b) 1 1: The SAR value is decremented after a data transfer. (–1 when the SZ[1:0] bits are 00b, –2 when 01b, –4 when 10b)	—
b5, b4	SZ[1:0]	DTC Data Transfer Size	b5 b4 0 0: Byte (8-bit) transfer 0 1: Word (16-bit) transfer 1 0: Longword (32-bit) transfer 1 1: Setting prohibited	—
b7, b6	MD[1:0]	DTC Transfer Mode Select	b7 b6 0 0: Normal transfer mode 0 1: Repeat transfer mode 1 0: Block transfer mode 1 1: Setting prohibited	—

MRA register is used to select the DTC operating mode and cannot be accessed directly from the CPU.

WBDIS Bit (Write-back Disable)

The WBDIS bit selects whether to write back the transfer information.

When the bit is 0, updated transfer information is written back.

When the bit is 1, updated transfer information is not written back even with the setting of that address is incremented after a transfer, and the same data transfer is executed every time for each transfer request. The transfer information can be stored in ROM because the transfer information is not written back.

While the WBDIS bit is 1, operation for each transfer mode is as follows:

(1) Normal transfer and repeat transfer modes

1-byte, 1-word, or 1-longword of data is transferred on a single transfer request. The transfer address and transfer count are not updated so that the same transfer is repeated on each transfer request. When the transfer count is 1, the ICU.DTCERn.DTCE bit is not set to 0, and data transfer continues in response to the next transfer request.

(2) Block transfer mode

1-block of data is transferred on a single transfer request. The transfer address and transfer count are not updated so that the same block transfer is repeated on each transfer request. When the block transfer count is 1, the ICU.DTCERn.DTCE bit is not set to 0, and data transfer continues in response to the next transfer request.

When setting the DISPE bit to 1, set the MRA.WBDIS bit to 1 (does not write back the transfer information). If the value of the WBDIS bit in any transfer information is 1, set the DTCCR.RRS bit to 0 (so that reading of the transfer information is not skipped).

20.2.2 DTC Mode Register B (MRB)

Address(es): (inaccessible directly from the CPU)

	b7	b6	b5	b4	b3	b2	b1	b0
	CHNE	CHNS	DISEL	DTS	DM[1:0]		INDX	SQEND
Value after reset:	x	x	x	x	x	x	x	x

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	SQEND	Sequence Transfer End	0: Continue the sequence transfer 1: End the sequence transfer	—
b1	INDX	Index Table Reference	0: Does not refer to the index table 1: Refers the index table based on the transferred data*1	—
b3, b2	DM[1:0]	Transfer Destination Address Addressing Mode	b3 b2 0 0: The address in the DAR register is fixed. (Write-back to DAR is skipped.) 0 1: The address in the DAR register is fixed. (Write-back to DAR is skipped.) 1 0: The DAR value is incremented after data transfer. (+1 when the MRA.SZ[1:0] bits are 00b, +2 when 01b, +4 when 10b) 1 1: The DAR value is decremented after data transfer. (-1 when the MRA.SZ[1:0] bits are 00b, -2 when 01b, -4 when 10b)	—
b4	DTS	DTC Transfer Mode Select	0: Transfer destination side is repeat area or block area. 1: Transfer source side is repeat area or block area.	—
b5	DISEL	DTC Interrupt Select	0: An interrupt request to the CPU is generated on completion of the specified number of data transfers. 1: An interrupt request to the CPU is generated for each data transfer.	—
b6	CHNS	DTC Chain Transfer Select	0: Chain transfer is performed on completion of each transfer. 1: Chain transfer is performed only when the transfer counter is changed from 1 to 0 or 1 to CRAH.	—
b7	CHNE	DTC Chain Transfer Enable	0: Chain transfer is disabled. 1: Chain transfer is enabled.	—

Note 1. Set the MRA.MD[1:0] bits to 00b (normal transfer mode) when setting the INDX bit to 1.

MRB register is used to select the DTC operating mode and cannot be accessed directly from the CPU.

SQEND Bit (Sequence Transfer End)

The SQEND bit selects whether to continue or end sequence transfer. Refer to Table 20.2 for details.

This bit can only be set to 1 for transfer information referred to by the DTC index table. Set this bit to 0 for transfer information referred to by the DTC vector table.

INDX Bit (Index Table Reference)

When the value of the INDX bit in transfer information that is read is 1, a sequence transfer proceeds. Refer to Table 20.2 for details.

Set this bit to 0 for transfer information which is not associated with sequence transfer or is not intended to start sequence transfer. Do not allow transfer requests to be generated by the sources different from that specified in the DTCSQE register but having the INDX bit set to 1.

Table 20.2 Values of Bits CHNE, SQEND, and INDX in the Sequence Transfer and DTC Operation

CHNE Bit	SQEND Bit	INDX Bit	Operation	Usage
0	0	1	Start sequence transfer	Use this setting for the transfer information that is first read in response to a transfer request from the source specified in the DTCSQE register.
1	0	0	Continue sequence transfer	Use this setting for the first or intermediate transfer information in a sequence.
0	0	0	Suspend sequence transfer	Use this setting for the first or intermediate transfer information in a sequence.
0	1	0	End sequence transfer	Use this setting with the last transfer information in a sequence.
0	1	1	End current sequence transfer and start new sequence transfer	Use this setting with the last transfer information in a sequence.

Note: Do not set the values other than listed above.

DTS Bit (DTC Transfer Mode Select)

The DTS bit specifies the side (transfer source or destination) to be a repeat area or block area in repeat transfer mode or block transfer mode.

CHNS Bit (DTC Chain Transfer Select)

The CHNS bit selects the chain transfer condition.

When the CHNE bit is 0, setting of the CHNS bit is ignored. For details on the conditions to select the chain transfer, refer to Table 20.4, Chain Transfer Conditions.

When the next transfer is chain transfer, completion of the specified number of transfers is not determined, the interrupt status flag for the request source is not cleared, and an interrupt request to the CPU is not generated.

CHNE Bit (DTC Chain Transfer Enable)

The CHNE bit enables or disables chain transfer.

The chain transfer condition is selected by the CHNS bit.

For details of chain transfer, refer to section 20.4.6, Chain Transfer.

Refer to Table 20.2 for the setting value to be used in the sequence transfer.

20.2.3 DTC Mode Register C (MRC)

Address(es): (inaccessible directly from the CPU)

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	DISPE
Value after reset:	x	x	x	x	x	x	x	x

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	DISPE	Displacement Addition	0: The displacement value is not added to the transfer source address. 1: The displacement value is added to the transfer source address.	—
b7 to b1	—	Reserved	Set these bits to 0.	—

The MRC register is used to select DTC operating mode and cannot be accessed directly from the CPU.

This register can only be used in full-address mode, but not in short-address mode. Therefore, set the DTCADM.SHORT bit to 0 (full-address mode) when using the displacement addition function.

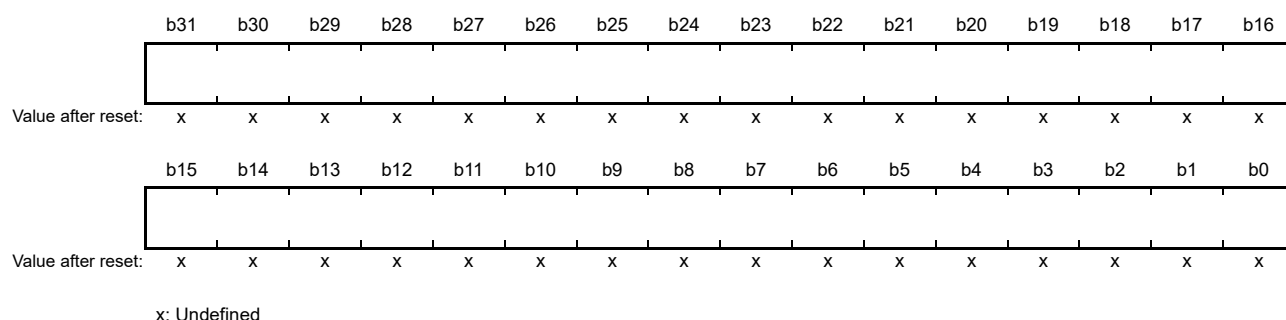
DISPE Bit (Displacement Addition)

This bit specifies whether to use the SAR + DTCDISP value as the transfer source address.

When setting the DISPE bit to 1, set the MRA.WBDIS bit to 1 (does not write back the transfer information) and set the DTCCR.RRS bit to 0 (transfer information read is not skipped).

20.2.4 DTC Transfer Source Register (SAR)

Address(es): (inaccessible directly from the CPU)



SAR register is used to set the transfer source start address.

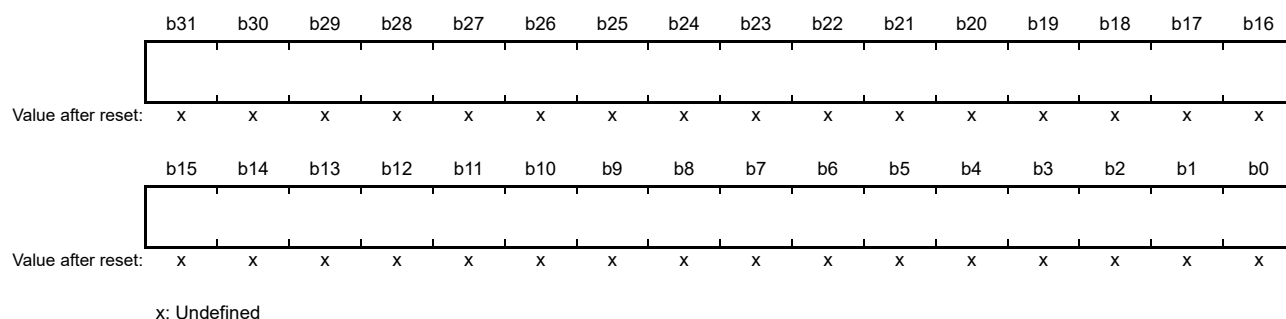
In full-address mode, 32 bits are valid.

In short-address mode, lower 24 bits are valid and upper 8 bits (b31 to b24) are ignored. The address of this register is extended by the value specified by b23.

SAR register cannot be accessed directly from the CPU.

20.2.5 DTC Transfer Destination Register (DAR)

Address(es): (inaccessible directly from the CPU)



DAR register is used to set the transfer destination start address.

In full-address mode, 32 bits are valid.

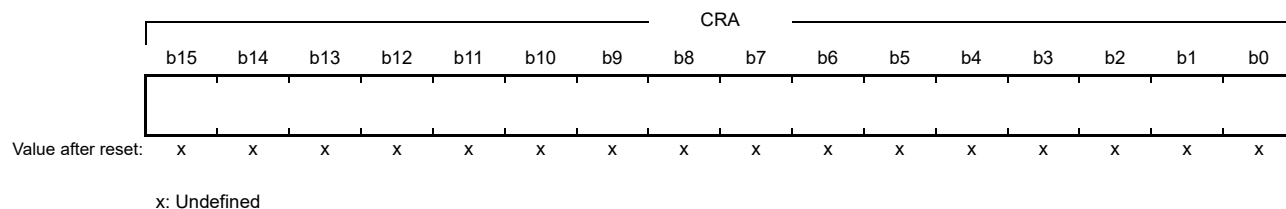
In short-address mode, lower 24 bits are valid and upper 8 bits (b31 to b24) are ignored. The address of this register is extended by the value specified by b23.

DAR register cannot be accessed directly from the CPU.

20.2.6 DTC Transfer Count Register A (CRA)

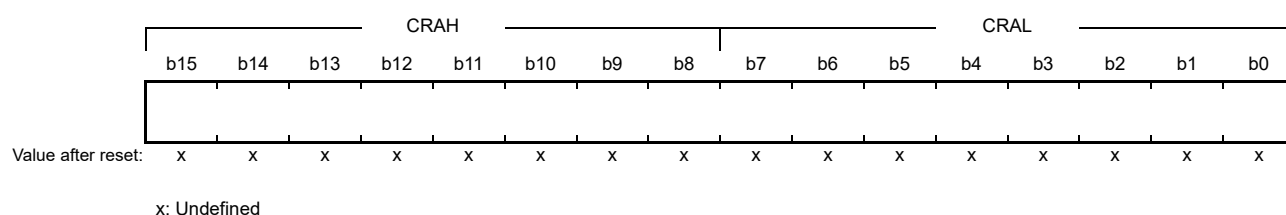
- Normal transfer mode

Address(es): (inaccessible directly from the CPU)



- Repeat transfer mode/block transfer mode

Address(es): (inaccessible directly from the CPU)



Symbol	Register Name	Description	R/W
CRAL	Transfer Counter A Lower Register	Set transfer count. This register functions as a transfer counter during data transfer.	—
CRAH	Transfer Counter A Upper Register	Set transfer count. This register functions as a reload register during data transfer.	—

Note: The function depends on transfer mode.

Note: Set CRAH and CRAL to the same value in repeat transfer mode and block transfer mode.

This register is for counting the number of transfers and cannot be accessed directly from the CPU.

(1) Normal transfer mode (MRA.MD[1:0] bits = 00b)

CRA register functions as a 16-bit transfer counter in normal transfer mode.

The transfer count is 1, 65535, and 65536 when the set value is 0001h, FFFFh, and 0000h, respectively.

The CRA value is decremented (−1) at each data transfer.

(2) Repeat transfer mode (MRA.MD[1:0] bits = 01b)

The CRAH register retains the transfer count and the CRAL register functions as an 8-bit transfer counter.

The transfer count is 1, 255, and 256 when the set value is 01h, FFh, and 00h, respectively.

The CRAL value is decremented (−1) at each data transfer. When it reaches 00h, the CRAH value is reloaded to the CRAL register.

(3) Block transfer mode (MRA.MD[1:0] bits = 10b)

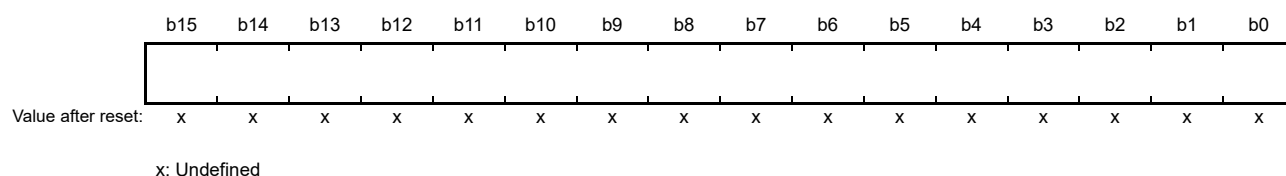
The CRAH register retains the block size and the CRAL register functions as an 8-bit block size counter.

The transfer count is 1, 255, and 256 when the set value is 01h, FFh, and 00h, respectively.

The CRAL value is decremented (−1) at each data transfer. When it reaches 00h, the CRAH value is reloaded to the CRAL register.

20.2.7 DTC Transfer Count Register B (CRB)

Address(es): (inaccessible directly from the CPU)



CRB register is used to set the block transfer count for block transfer mode and cannot be accessed directly from the CPU.

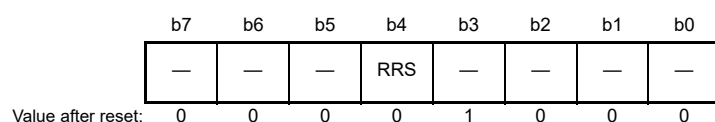
The transfer count is 1, 65535, and 65536 when the set value is 0001h, FFFFh, and 0000h, respectively.

The CRB value is decremented (–1) when the final data of a single block size is transferred.

When normal transfer mode or repeat transfer mode is selected, this register is not used and the set value is ignored.

20.2.8 DTC Control Register (DTCCR)

Address(es): DTC.DTCCR 0008 2400h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b3	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b4	RRS	DTC Transfer Information Read Skip Enable*1	0: Transfer information read is not skipped. 1: Transfer information read is skipped when vector numbers match.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Set this bit to 0 when using the sequence transfer.

DTCCR register is used to control the DTC operation.

RRS Bit (DTC Transfer Information Read Skip Enable)

The DTC vector number is compared with the vector number in the previous data transfer.

When these vector numbers match and the RRS bit is set to 1, DTC data transfer is performed without reading the transferred information. However, when the previous transfer was chain transfer, the transferred information is read regardless of the value of the RRS bit.

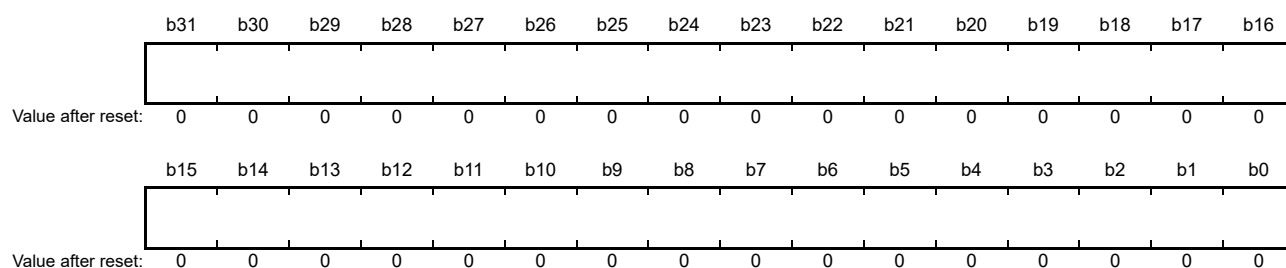
Furthermore, when the transfer counter (CRA register) became 0 during the previous normal transfer and when the transfer counter (CRB register) became 0 during the previous block transfer, the transferred information is read regardless of the RRS bit value.

If the value of the MRA.WBDIS bit in any transfer information is 1, set the RRS bit to 0. Note that the MRA.WBDIS bit should be set to 1 when the MRC.DISPE bit is set to 1.

Like chain transfer, sequence transfer handles sequences of multiple types of data transfer. When sequence transfer is to be used, set the RRS bit to 0 so that the previous data transfer will not be repeated.

20.2.9 DTC Vector Base Register (DTCVBR)

Address(es): DTC.DTCVBR 0008 2404h

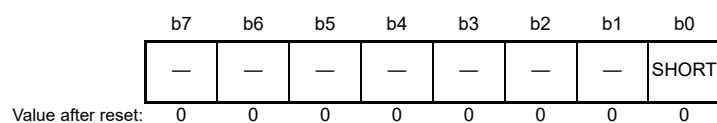


The DTCVBR register is used to set the base address for calculating the address to which the DTC vector is allocated. Values for the upper 4 bits (b31 to b28) cannot be written but reflect the value written to b27. The lower 10 bits are reserved and the values are fixed to 0. Write 0 to the lower 10 bits if necessary.

It can be set in the range of 0000 0000h to 07FF FC00h and F800 0000h to FFFF FC00h in 1-Kbyte units.

20.2.10 DTC Address Mode Register (DTCADMOD)

Address(es): DTC.DTCADMOD 0008 2408h



Bit	Symbol	Bit Name	Description	R/W
b0	SHORT	Short-Address Mode Set*1	0: Full-address mode 1: Short-address mode	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Set this bit to 0 (full-address mode) when using the sequence transfer.

DTCADMOD register is used to specify the area accessible by the DTC.

SHORT Bit (Short-Address Mode Set)

This bit is used to select address mode of registers SAR and DAR.

Full-address mode allows the DTC to access to a 4-Gbyte space (0000 0000h to FFFF FFFFh).

Short-address mode allows the DTC to access to a 16-Mbyte space (0000 0000h to 007F FFFFh and FF80 0000h to FFFF FFFFh).

20.2.11 DTC Module Start Register (DTCST)

Address(es): DTC.DTCST 0008 240Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	DTCST
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DTCST	DTC Module Start	0: DTC module stop 1: DTC module start	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

DTCST Bit (DTC Module Start)

Set the DTCST bit to 1 to enable the DTC to accept transfer requests. When this bit is set to 0, transfer requests are no longer accepted.

If this bit is set to 0 during data transfer, the accepted transfer request is active until the processing is completed.

Set the DTCST bit to 0 before making a transition to the module stop state, all-module clock stop mode, software standby mode, or deep software standby mode.

Set the DTCST bit to 1 to resume the data transfer after returning from the module stop state, all-module clock stop mode, or software standby mode.

For details on transitions to the module stop state, all-module clock stop mode, software standby mode, and deep software standby mode, refer to section 20.9, Low Power Consumption Function, and section 11, Low Power Consumption.

20.2.12 DTC Status Register (DTCSTS)

Address(es): DTC.DTCSTS 0008 240Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ACT	—	—	—	—	—	—	—	VECN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	VECN[7:0]	DTC Active Vector Number Monitoring Flag	These bits indicate the vector number for the request source when data transfer is in progress. The value is only valid if data transfer is in progress (the value of the ACT flag is 1).	R
b14 to b8	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b15	ACT	DTC Active Flag	0: Data transfer is not in progress. 1: Data transfer is in progress.	R

VECN[7:0] Flags (DTC Active Vector Number Monitoring Flag)

While data transfer is in progress, these bits indicate the vector number corresponding to the request source for the transfer.

When the DTCSTS register is read, the value of the VECN[7:0] flags is valid if the value of the ACT flag was 1 (data transfer is in progress) and invalid if the value of the ACT flag was 0 (data transfer is not in progress).

For the correspondence between the DTC request sources and the vector addresses, refer to Table 15.5, Interrupt Vector Table in section 15, Interrupt Controller (ICUD).

ACT Flag (DTC Active Flag)

This flag indicates the state of data transfer operation.

[Setting condition]

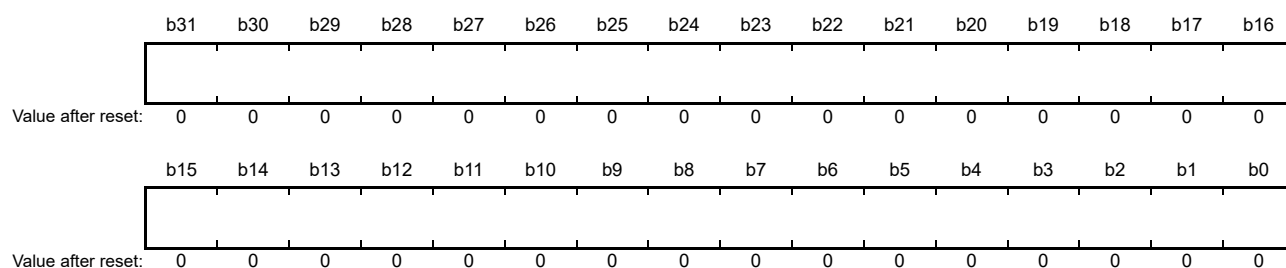
- When the data transfer is started by a transfer request.
- When the sequence transfer is resumed.

[Clearing condition]

- When the data transfer is completed in response to a transfer request.
- When the sequence transfer is suspended.

20.2.13 DTC Index Table Base Register (DTCIBR)

Address(es): DTC.DTCIBR 0008 2410h



The DTCIBR register is used to set the base address for calculating the address to which the DTC index is allocated.

Values for the upper 4 bits (b31 to b28) cannot be written but reflect the value written to b27.

The lower 10 bits (b9 to b0) are reserved bits and fixed to 0. When writing this register, set these bits to 0.

It can be set in the range of 0000 0000h to 07FF FC00h and F800 0000h to FFFF FC00h in 1-Kbyte units.

20.2.14 DTC Operation Register (DTCOR)

Address(es): DTC.DTCOR 0008 2414h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	SQTFRL
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SQTFRL	Sequence Transfer Terminate	Writing 1 to this bit terminates the sequence transfer in progress. This bit is read as 0.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

The DTCOR register sets the operation of the DTC module.

SQTFRL Bit (Sequence Transfer Terminate)

Setting the SQTFRL bit to 1 terminates the sequence transfer in progress.

When the DTCSQE.ESPSEL bit is 1 (Sequence transfer is enabled), follow the procedure shown in Figure 20.2 to terminate the sequence transfer.

Writing 1 to the bit, while no sequence transfer is performed, have no effect.

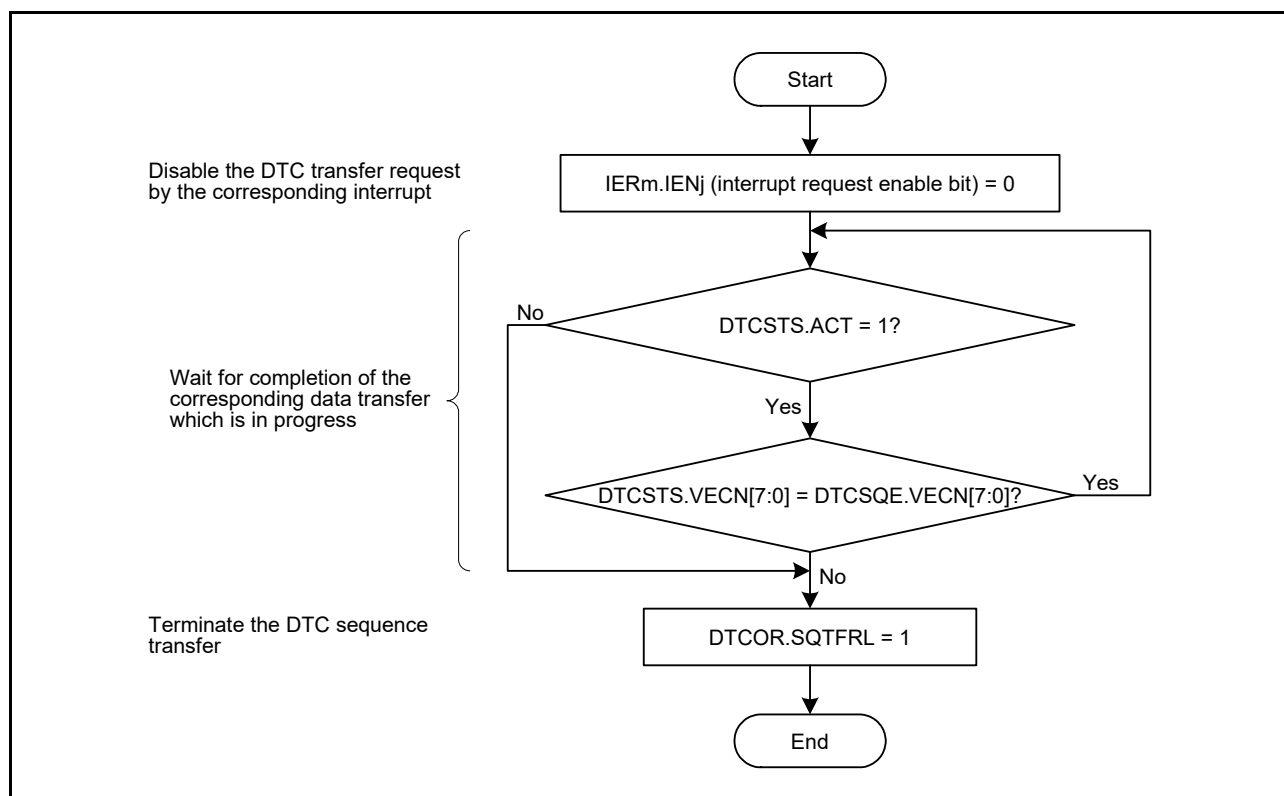


Figure 20.2 Procedure to Terminate Sequence Transfer

20.2.15 DTC Sequence Transfer Enable Register (DTCSQE)

Address(es): DTC.DTCSQE 0008 2416h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	ESPSEL	—	—	—	—	—	—	—	VECN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	VECN[7:0]	Sequence Transfer Vector Number Setting	Specify the vector number by which a sequence transfer is enabled. The value is only valid when the ESPSEL bit is 1.	R/W
b14 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R
b15	ESPSEL	Sequence Transfer Enable	0: Sequence transfer is disabled. 1: Sequence transfer is enabled.	R/W

The DTCSQE register is used to specify sequence transfer. Follow Figure 20.24 for details on the setting procedure.

VECN[7:0] Bit (Sequence Transfer Vector Number Setting)

This bit is used to specify for which vector number to perform sequence transfer. Sequence transfer can occur only for this trigger source.

Table 15.5, Interrupt Vector Table in section 15, Interrupt Controller (ICUD) shows the relationship between the trigger source and the vector number.

ESPSEL Bit (Sequence Transfer Enable)

The ESPSEL bit specifies whether sequence transfer is used.

Set the DTCADMOD.SHORT bit to 0 (full address mode), when setting the ESPSEL bit to 1.

20.2.16 DTC Address Displacement Register (DTCDISP)

Address(es): DTC.DTCDISP 0008 2418h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The DTCDISP register is used to specify the displacement value to add to the DTC transfer source address.

If MRC.DISPE bit is 1, the value SAR + DTCDISP is used as the transfer source address.

20.3 Request Sources

The DTC data transfer is triggered by an interrupt request. Setting the ICU.DTCERn.DTCE bit (n = interrupt vector number) to 1 selects the corresponding interrupt request as a request source for the DTC.

For the correspondence between the DTC request sources and the vector addresses, refer to Table 15.5, Interrupt Vector Table in section 15, Interrupt Controller (ICUD). For request by software, refer to section 15.2.5, Software Interrupt Generation Register (SWINTR) and section 15.2.6, Software Interrupt 2 Generation Register (SWINT2R) in section 15, Interrupt Controller (ICUD).

Once the DTC has accepted a transfer request, it does not accept another transfer request until transfer for that single request is completed, regardless of the priority of the requests.

When multiple transfer requests are generated during data transfer by the DMAC/DTC, the request with the highest priority on completion of the current transfer is accepted. When multiple transfer requests are generated while the DTCST.DTCST bit is 0 (DTC module stop), the request with the highest priority at the moment when the bit is subsequently set to 1 (DTC module start) is accepted.

The DTC performs the following operations at the start of a single data transfer (or the last of the consecutive transfers in the case of a chain transfer).

- On completion of a specified number of data transfer, the ICU.DTCERn.DTCE bit is set to 0 and an interrupt is requested to the CPU.
- If the MRB.DISEL bit is 1, an interrupt is requested to the CPU on completion of data transfer.
- For the other transfers, the interrupt status flag of the request source is set to 0 at the start of data transfer.

20.3.1 Allocating Transfer Information and DTC Vector Table

The DTC reads the start address of the transfer information corresponding to each request source from the vector table and reads the transfer information starting at that address.

The vector table should be located so that the lower 10 bits of the base address (start address) are 0. Use the DTC vector base register (DTCVBR) to set the base address of the DTC vector table.

Transfer information is allocated in the RAM area. Transfer information can be allocated in the ROM area when the MRA.WBDIS bit is set to 1. The start address of the transfer information n with vector number n should be allocated at DTCVBR + 4n.

Transfer information should be aligned on a 4-byte boundary. The size of a transfer information is 12 bytes in short-address mode or 16 bytes in full-address mode. Use the DTCADMOD.SHORT bit to select short-address mode (SHORT bit = 1) or full-address mode (SHORT bit = 0).

Figure 20.3 shows the relationship between the DTC vector table and transfer information.

Figure 20.4 shows the allocation of transfer information in the RAM area. The lower addresses vary according to the endian of the corresponding allocation area. For details, refer to section 20.10.2, Allocating Transfer Information.

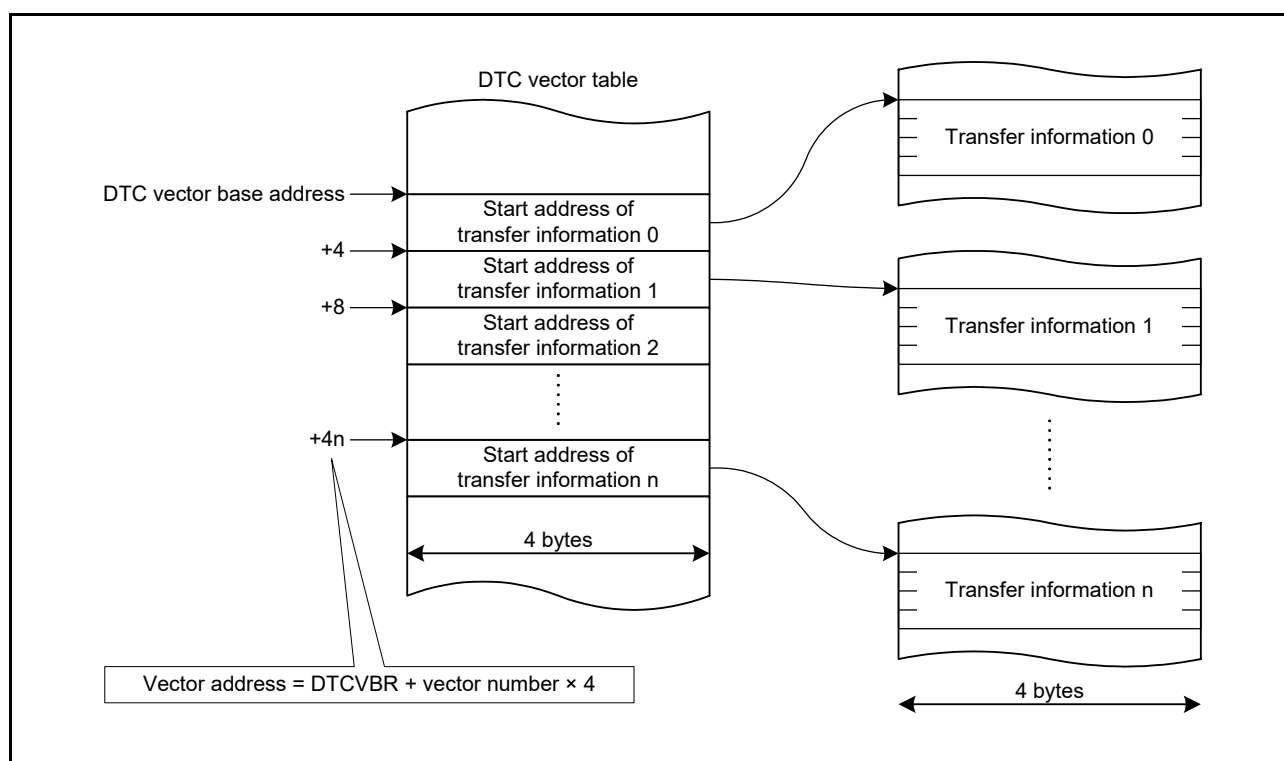


Figure 20.3 DTC Vector Table and Transfer Information

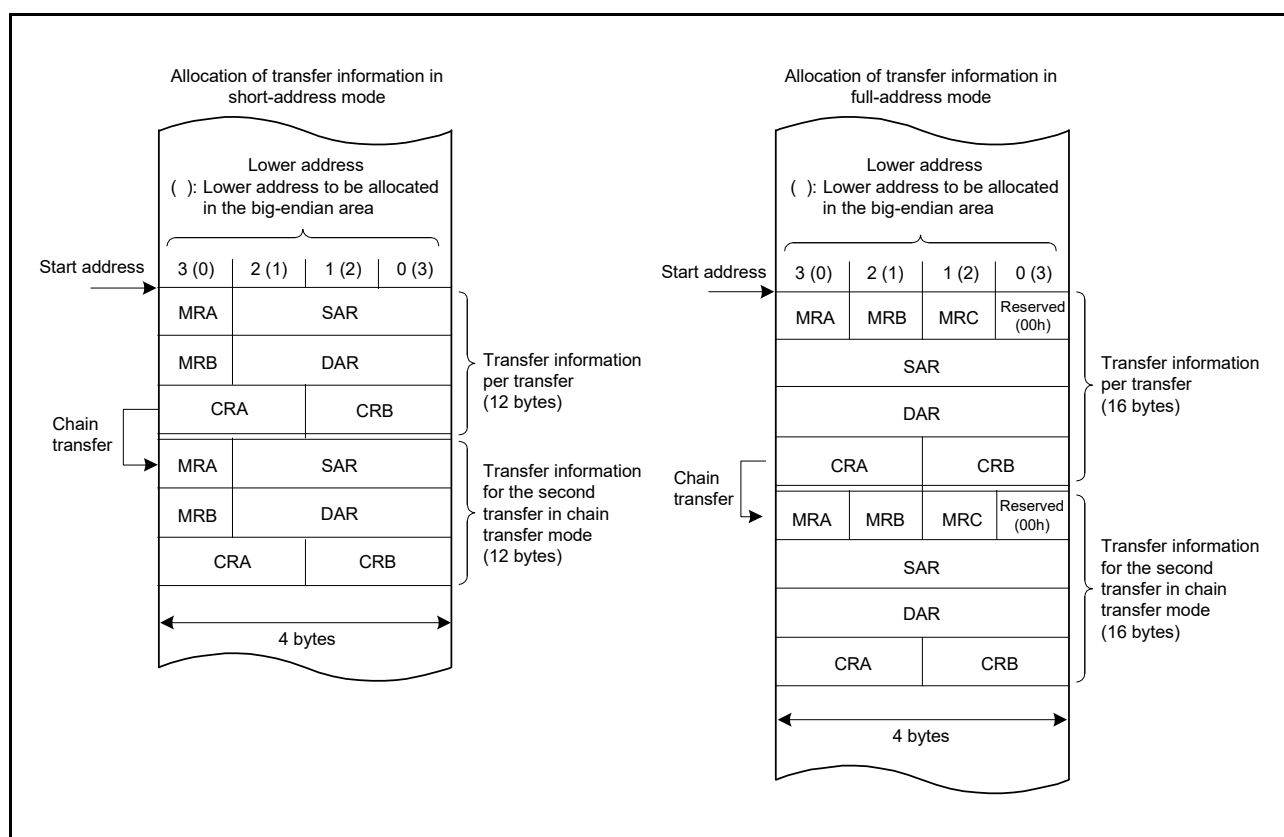


Figure 20.4 Allocation of Transfer Information in the RAM Area

20.4 Operation

The DTC transfers data in accordance with the transfer information. Storage of the transfer information in the RAM area is required before DTC operation.

When the DTC accepts a transfer request, it reads the DTC vector corresponding to the vector number. Next, the DTC reads transfer information from the address pointed by the DTC vector, transfers data, and then writes back the transfer information after the data transfer. Allocating transfer information in the RAM area allows data transfer of arbitrary number of channels.

There are three transfer modes: normal transfer mode, repeat transfer mode, and block transfer mode.

Set a transfer source address in the SAR register and a transfer destination address in the DAR register. The SAR and DAR registers are updated after the transfer according to the respective settings (increment, decrement, or fixed).

Table 20.3 lists transfer modes of the DTC.

Table 20.3 Transfer Modes of the DTC

Transfer Mode	Data Size Transferred on Single Transfer Request	Increment/Decrement of Memory Address	Settable Transfer Count
Normal transfer mode	1 byte/1 word/1 longword	Incremented/decremented by 1, 2, or 4 or address fixed	1 to 65536
Repeat transfer mode*1	1 byte/1 word/1 longword	Incremented/decremented by 1, 2, or 4 or address fixed	1 to 256*3
Block transfer mode*2	Block size specified in CRAH (1 to 256 bytes/1 to 256 words/1 to 256 longwords)	Incremented/decremented by 1, 2, or 4 or address fixed	1 to 65536

Note 1. Set transfer source or transfer destination in the repeat area.

Note 2. Set transfer source or transfer destination in the block area.

Note 3. After data transfer of the specified count, the initial state is restored and the operation is continued (repeated).

Setting the MRB.CHNE bit to 1 allows multiple transfers (chain transfer) on a single transfer request. The setting in combination with the MRB.CHNS bit enables a chain transfer when the specified number of data transfers is completed. Figure 20.5 shows the operation flowchart of the DTC. Table 20.4 lists chain transfer conditions.

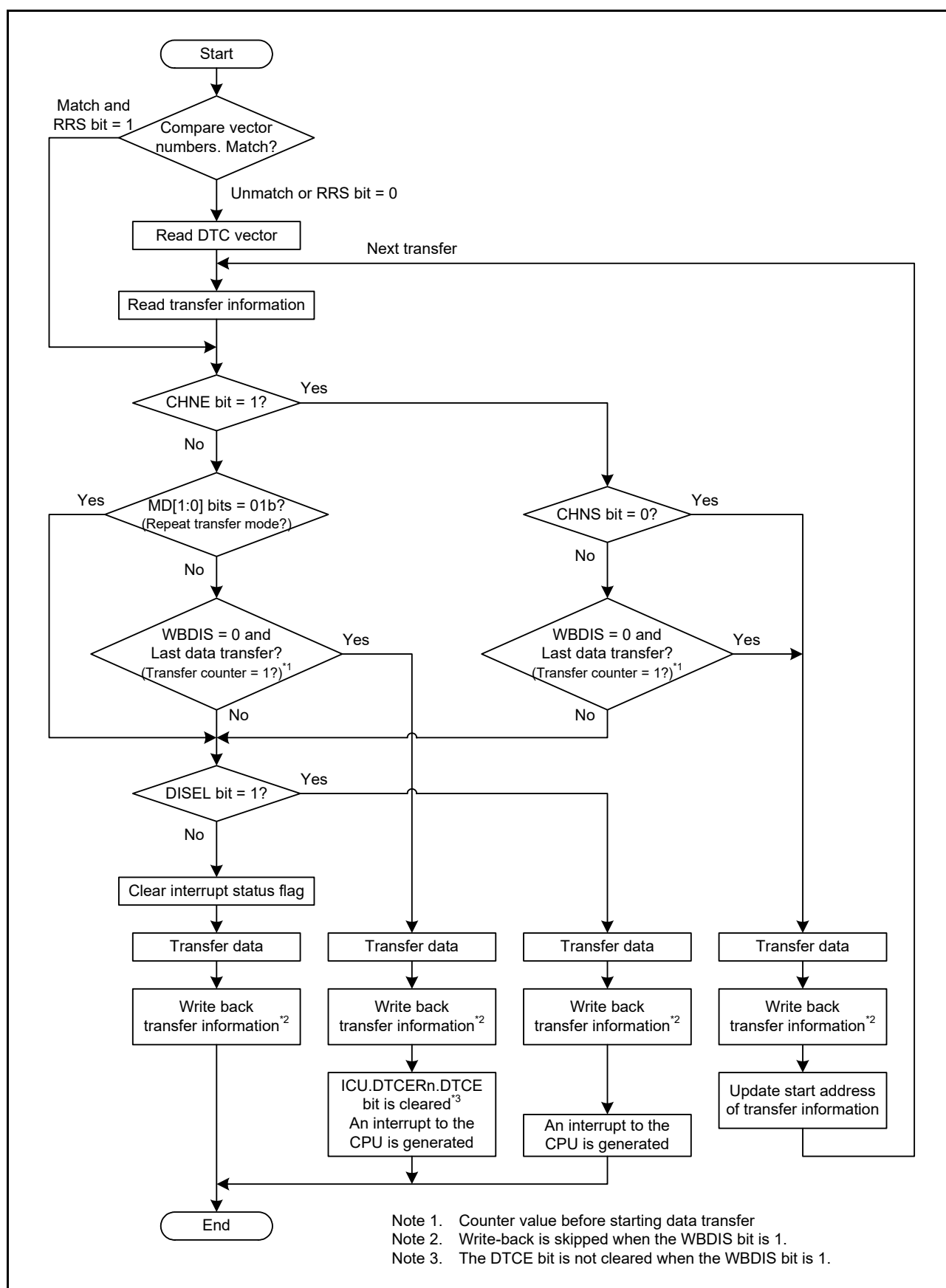


Figure 20.5 Operation Flowchart of the DTC

Table 20.4 Chain Transfer Conditions

First Transfer				Second Transfer*3				Data Transfer
CHNE Bit	CHNS Bit	DISEL Bit	Transfer Counter*1,*2	CHNE Bit	CHNS Bit	DISEL Bit	Transfer Counter*1,*2	
0	—	0	Other than (1 → 0)	—	—	—	—	Ends after the first transfer
0	—	0	(1 → 0)	—	—	—	—	Ends after the first transfer with an interrupt request to the CPU
0	—	1	—	—	—	—	—	
1	0	—	—	0	—	0	Other than (1 → 0)	Ends after the second transfer
				0	—	0	(1 → 0)	Ends after the second transfer with an interrupt request to the CPU
				0	—	1	—	
1	1	0	Other than (1 → *)	—	—	—	—	Ends after the first transfer
1	1	—	(1 → *)	0	—	0	Other than (1 → 0)	Ends after the second transfer
				0	—	0	(1 → 0)	Ends after the second transfer with an interrupt request to the CPU
				0	—	1	—	
1	1	1	Other than (1 → *)	—	—	—	—	Ends after the first transfer with an interrupt request to the CPU

Note 1. The transfer counters used depend on transfer modes as follows:

Normal transfer mode: CRA register

Repeat transfer mode: CRAL register

Block transfer mode: CRB register

Note 2. On completion of data transfer, the counters operate as follows:

1 → 0 in normal and block transfer modes

1 → CRAH in repeat transfer mode

(1 → *) in the table indicates both of the two operations above.

Note 3. Chain transfer can be selected for the second or subsequent transfers. The condition combination of “second transfer and the CHNE bit is 1” is omitted.

20.4.1 Transfer Information Read Skip Function

Reading of DTC vector and transfer information can be skipped by the setting of the DTCCR.RRS bit.

When a DTC transfer request is accepted, the current DTC vector number is compared with the DTC vector number in the previous data transfer. When these vector numbers match and the RRS bit is 1, the DTC does not read the DTC vector and transfer information, and transfers data according to the transfer information remained in the DTC.

However, when the previous transfer was chain transfer, the DTC vector and transfer information are read. Furthermore, when the transfer counter (CRA register) became 0 during the previous normal transfer and when the transfer counter (CRB register) became 0 during the previous block transfer, transfer information is read regardless of the value of the RRS bit. Figure 20.14 shows an example of transfer information read skip.

When updating the vector table and transfer information, set the RRS bit to 0, update the vector table and transfer information, and then set the RRS bit to 1. Setting the RRS bit to 0 discards the vector numbers retained in the DTC. The updated DTC vector table and transfer information are read in the next data transfer.

20.4.2 Transfer Information Write-Back Skip Function

20.4.2.1 Write-Back Skip by Fixing Addresses

When the MRA.SM[1:0] bits or the MRB.DM[1:0] bits are set to “address is fixed” (00b or 01b), a part of transfer information is not written back. This function is performed independently of the setting of short-address mode or full-address mode.

Table 20.5 lists transfer information write-back skip conditions and applicable registers. The CRA and CRB registers are written back independently of the setting of short-address mode or full-address mode.

Furthermore, in full-address mode, write-back of registers MRA, MRB, and MRC is skipped.

Table 20.5 Transfer Information Write-Back Skip Conditions and Applicable Registers

MRA.SM[1:0] Bits		MRB.DM[1:0] Bits		SAR Register	DAR Register
b3	b2	b3	b2		
0	0	0	0	Skip	Skip
0	0	0	1		
0	1	0	0		
0	1	0	1		
0	0	1	0	Skip	Write-back
0	0	1	1		
0	1	1	0		
0	1	1	1		
1	0	0	0	Write-back	Skip
1	0	0	1		
1	1	0	0		
1	1	0	1		
1	0	1	0	Write-back	Write-back
1	0	1	1		
1	1	1	0		
1	1	1	1		

20.4.2.2 Write-Back Skip by the MRA.WBDIS Bit

When the MRA.WBDIS bit is 1, the transfer information (SAR, DAR, CRA, and CRB) is not written back regardless of the settings of the transfer information.

The transfer information on the memory is not updated, data can be transferred by the DTC without copying the transfer information from ROM to RAM. Skipping a write-back reduces time for post-processing of the data transfer.

20.4.3 Normal Transfer Mode

This mode allows 1-byte, 1-word, or 1-longword data transfer on a single transfer request. The transfer count can be set to 1 to 65536.

Transfer source addresses and transfer destination addresses can be set to increment, decrement, or fixed independently. This mode enables an interrupt request to the CPU to be generated at the end of specified-count transfer.

Table 20.6 lists register functions in normal transfer mode, and Figure 20.6 shows the memory map of normal transfer mode.

Table 20.6 Register Functions in Normal Transfer Mode

Register	Description	Value Written Back by Writing Transfer Information*1
SAR	Transfer source address	Increment/decrement/fixed*2
DAR	Transfer destination address	Increment/decrement/fixed*2
CRA	Transfer counter A	CRA – 1
CRB	Transfer counter B	Not updated

Note 1. Write-back operation is skipped when the MRA.WBDIS bit is 1.

Note 2. Write-back operation is skipped when address is fixed.

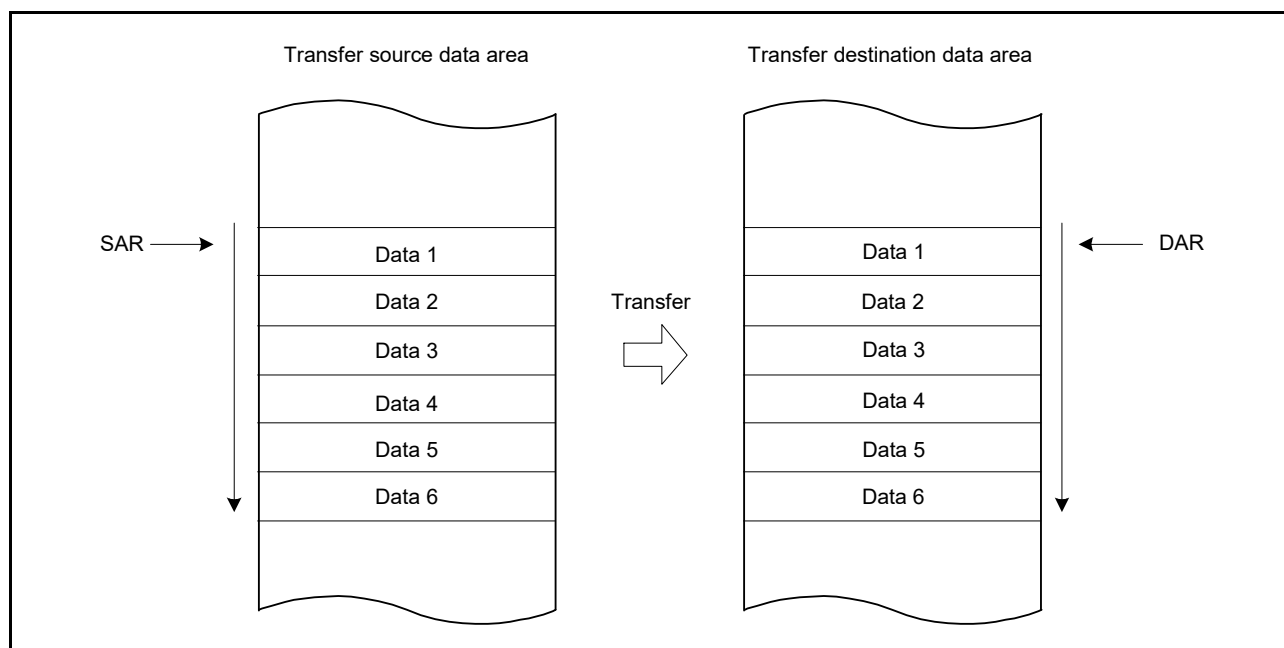


Figure 20.6 Memory Map of Normal Transfer Mode

20.4.4 Repeat Transfer Mode

This mode allows 1-byte, 1-word, or 1-longword data transfer on a single transfer request.

Specify either transfer source or transfer destination for the repeat area by the MRB.DTS bit. The transfer count can be set to 1 to 256. When the specified-count transfer is completed, the initial value of the address register specified in the transfer counter and the repeat area is restored and transfer is repeated. The other address register is incremented or decremented continuously or remains unchanged.

When the transfer counter CRAL is decreased to 00h in repeat transfer mode, the CRAL value is updated to the value set in the CRAH register. Thus the transfer counter does not become 00h, which disables an interrupt request to be generated to the CPU when the MRB.DISEL bit is set to 0 (an interrupt request to the CPU is generated on completion of the specified number of data transfers).

Table 20.7 lists the register functions in repeat transfer mode, and Figure 20.7 shows the memory map of repeat transfer mode.

Table 20.7 Register Functions in Repeat Transfer Mode

Register	Description	Value Written Back by Writing Transfer Information*1		
		When CRAL ≠ 1	When CRAL = 1	
			When the MRB.DTS Bit is 0	When the MRB.DTS Bit is 1
SAR	Transfer source address	Increment/decrement/fixed*2	Increment/decrement/fixed*2	SAR register initial value
DAR	Transfer destination address	Increment/decrement/fixed*2	DAR register initial value	Increment/decrement/fixed*2
CRAH	Retains initial value of transfer counter	CRAH	CRAH	
CRAL	Transfer counter A	CRAL – 1	CRAH	
CRB	Transfer counter B	Not updated	Not updated	

Note 1. Write-back operation is skipped when the MRA.WBDIS bit is 1.

Note 2. Write-back operation is skipped when address is fixed.

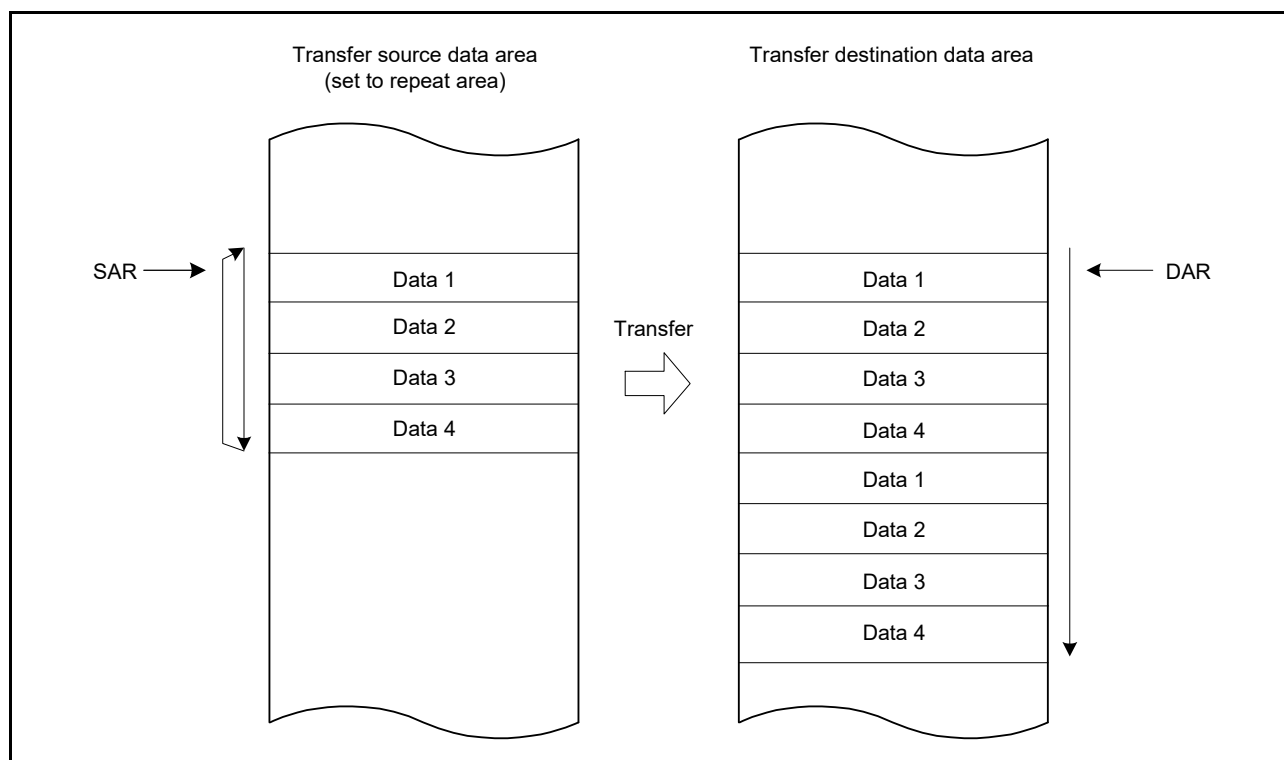


Figure 20.7 Memory Map of Repeat Transfer Mode (Transfer Source: Repeat Area)

20.4.5 Block Transfer Mode

This mode allows single-block data transfer on a single transfer request.

Specify either transfer source or transfer destination for the block area by the MRB.DTS bit. The block size can be set to 1 to 256 bytes, 1 to 256 words, or 1 to 256 longwords.

When transfer of the specified one block is completed, the initial values of the block size counter CRAL and the address register (the SAR register when the MRB.DTS bit is 1 or the DAR register when the DTS bit is 0) specified in the block area are restored. The other address register is incremented or decremented continuously or remains unchanged.

The transfer count (block count) can be set to 1 to 65536. This mode enables an interrupt request to the CPU to be generated at the end of specified-count block transfer.

Table 20.8 lists register functions in block transfer mode, and Figure 20.8 shows the memory map of block transfer mode.

Table 20.8 Register Functions in Block Transfer Mode

Register	Description	Value Written Back by Writing Transfer Information*1	
		When MRB.DTS Bit is 0	When MRB.DTS Bit is 1
SAR	Transfer source address	Increment/decrement/fixed*2	SAR register initial value
DAR	Transfer destination address	DAR register initial value	Increment/decrement/fixed*2
CRAH	Retains initial value of block size	CRAH	
CRAL	Block size counter	CRAH	
CRB	Block transfer counter	CRB – 1	

Note 1. Write-back operation is skipped when the MRA.WBDIS bit is 1.

Note 2. Write-back operation is skipped when address is fixed.

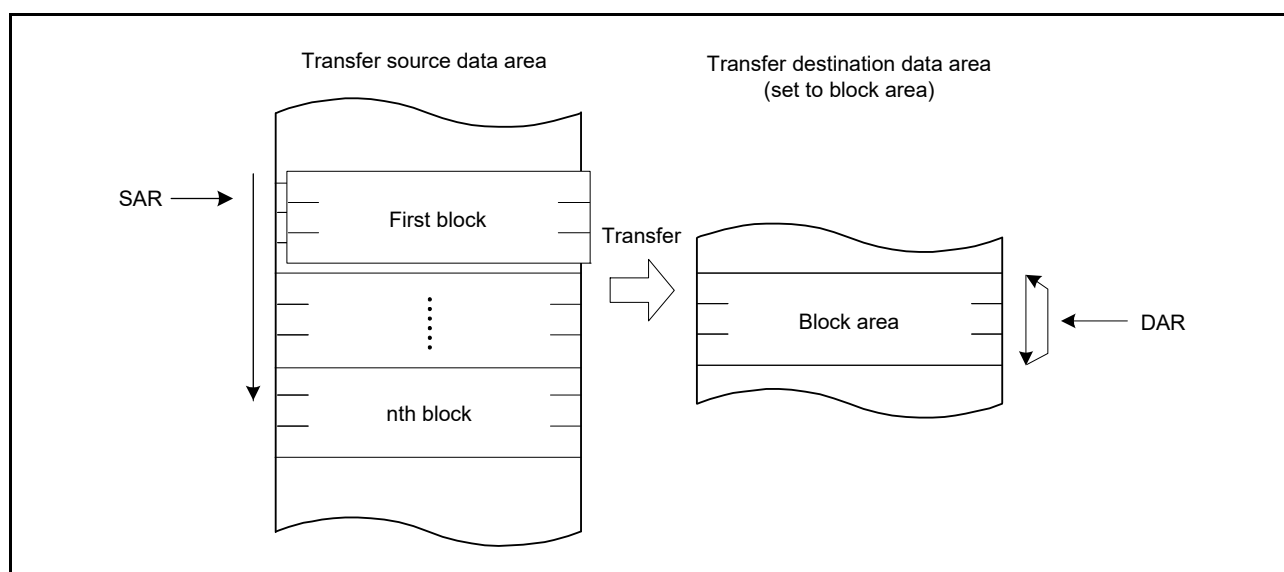


Figure 20.8 Memory Map of Block Transfer Mode (Transfer Destination: Block Area)

20.4.6 Chain Transfer

Setting the MRB.CHNE bit to 1 allows chain transfer to be performed continuously on a single transfer request.

If the MRB.CHNE bit is 1 and the MRB.CHNS bit is 0, an interrupt request to the CPU is not generated when the specified number of data transfers is completed, or while the MRB.DISEL bit is 1 (an interrupt request to the CPU is generated for every data transfer). Data transfer has no effect on the interrupt status flag, which is the request source.

The transfer information (SAR, DAR, CRA, CRB, MRA, MRB, and MRC) that define a data transfer can be specified independently of each other. Figure 20.9 shows chain transfer operation.

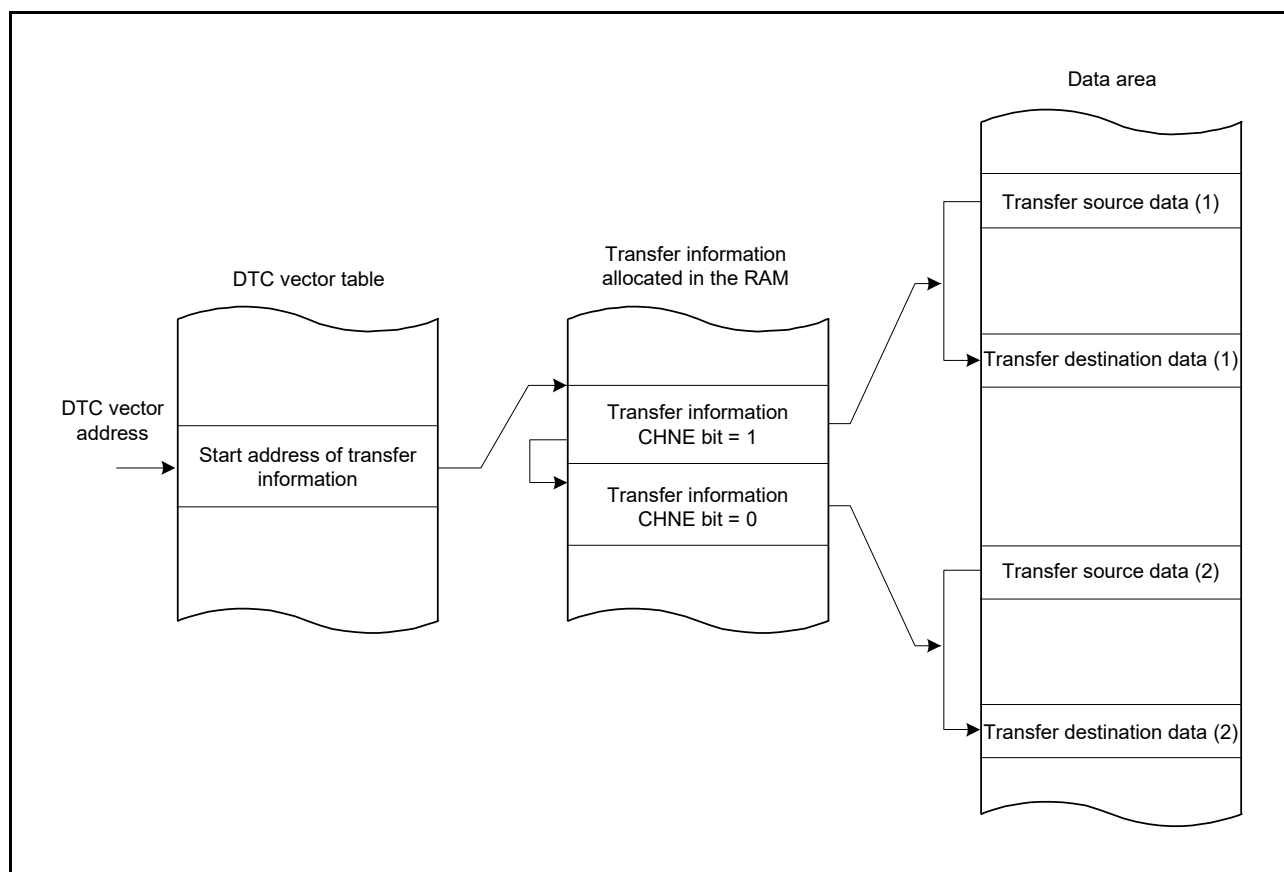


Figure 20.9 Chain Transfer Operation

If the MRB.CHNE bit is 1 and the CHNS bit is 1, chain transfer is performed only after completion of specified number of data transfers. In repeat transfer mode, chain transfer is performed after completion of specified number of data transfers.

For details on chain transfer conditions, refer to Table 20.4, Chain Transfer Conditions.

20.4.7 Operation Timing

Figure 20.10 to Figure 20.14 show examples of DTC operation timing.

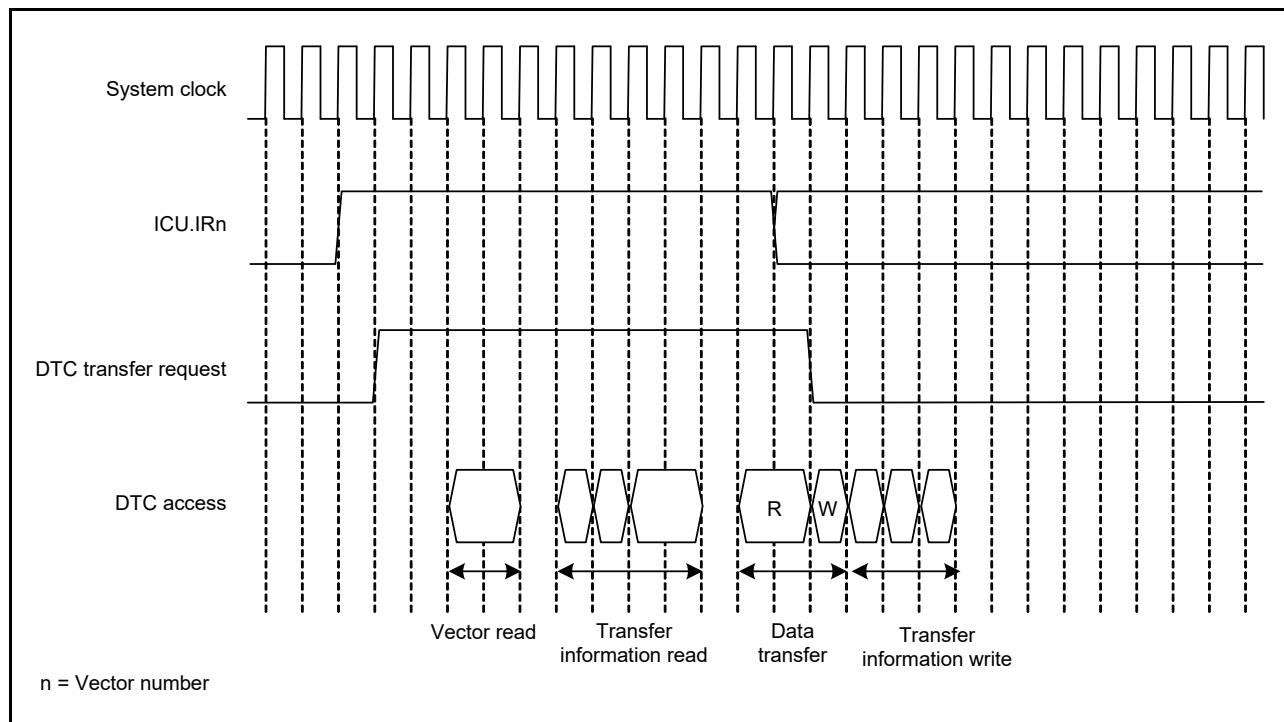


Figure 20.10 Example (1) of DTC Operation Timing
(Short-Address Mode, Normal Transfer Mode, Repeat Transfer Mode)

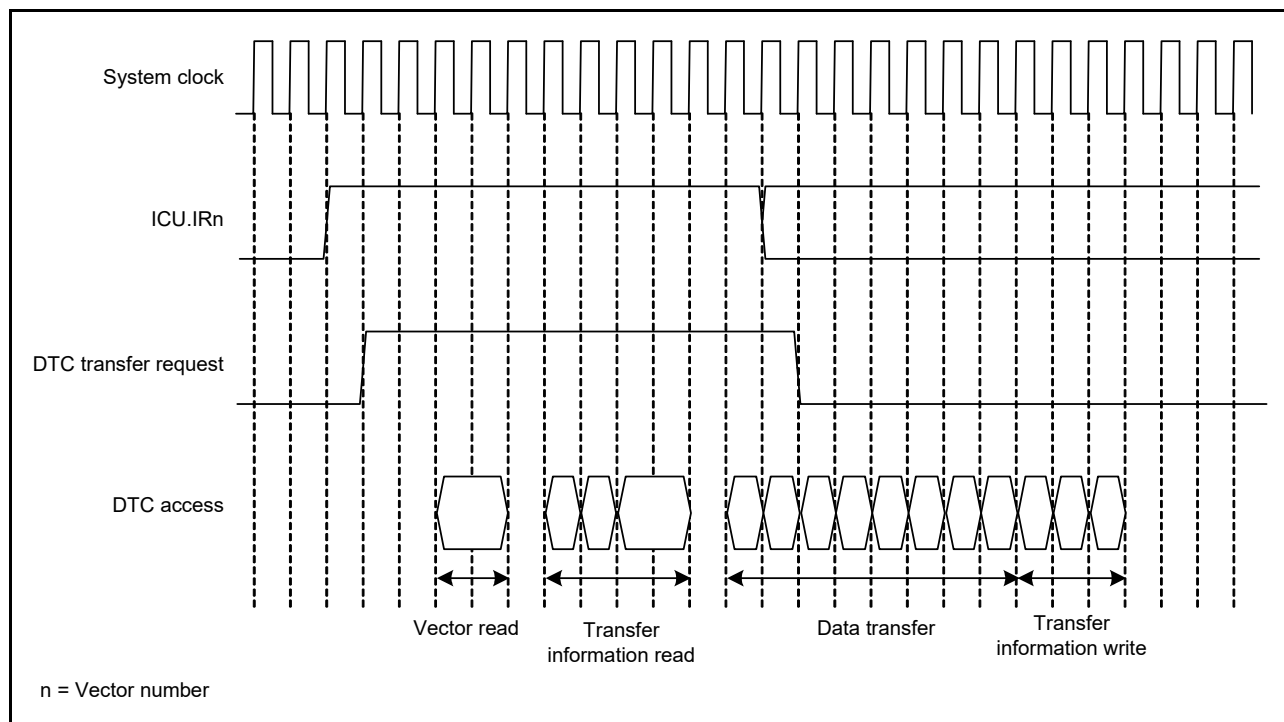


Figure 20.11 Example (2) of DTC Operation Timing
(Short-Address Mode, Block Transfer Mode, Block Size = 4)

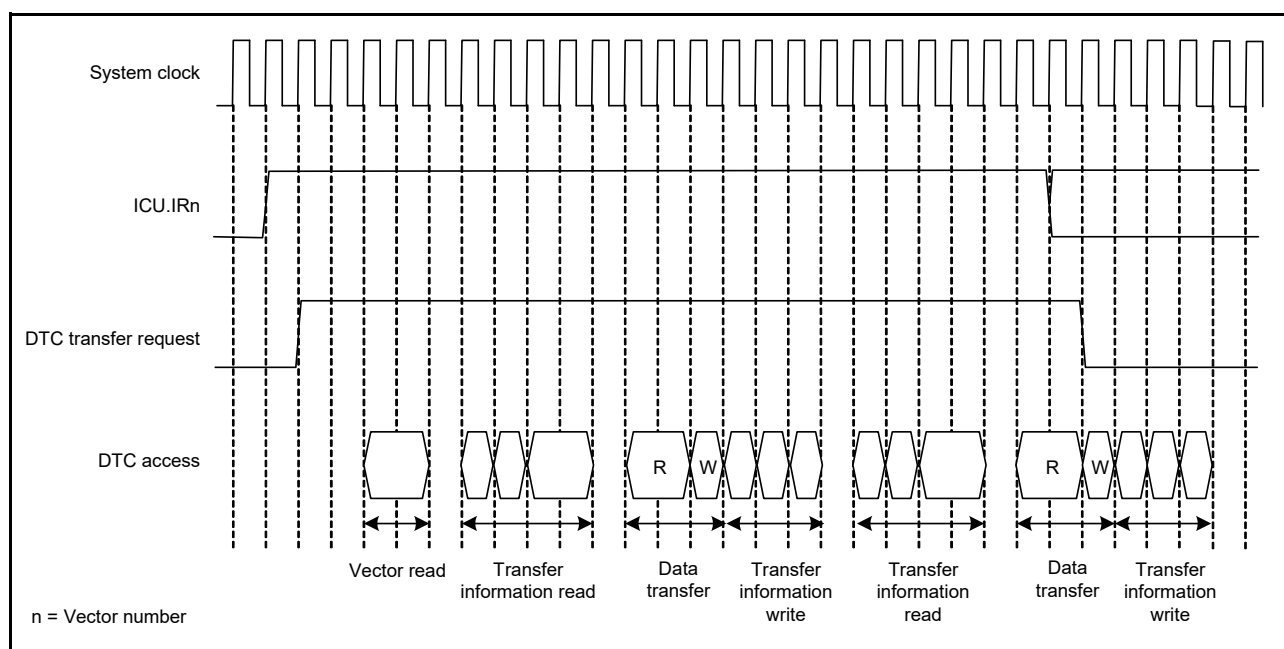


Figure 20.12 Example (3) of DTC Operation Timing (Short-Address Mode, Chain Transfer)

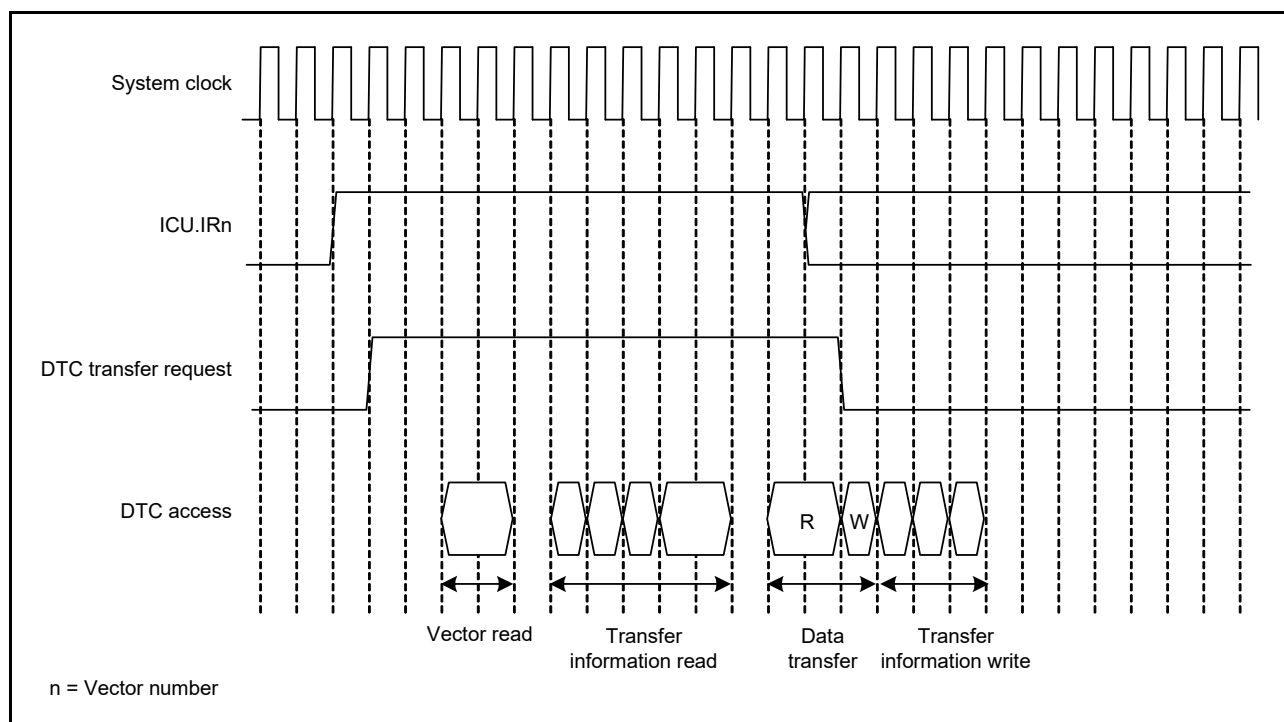


Figure 20.13 Example (4) of DTC Operation Timing (Full-Address Mode, Normal Transfer Mode, Repeat Transfer Mode)

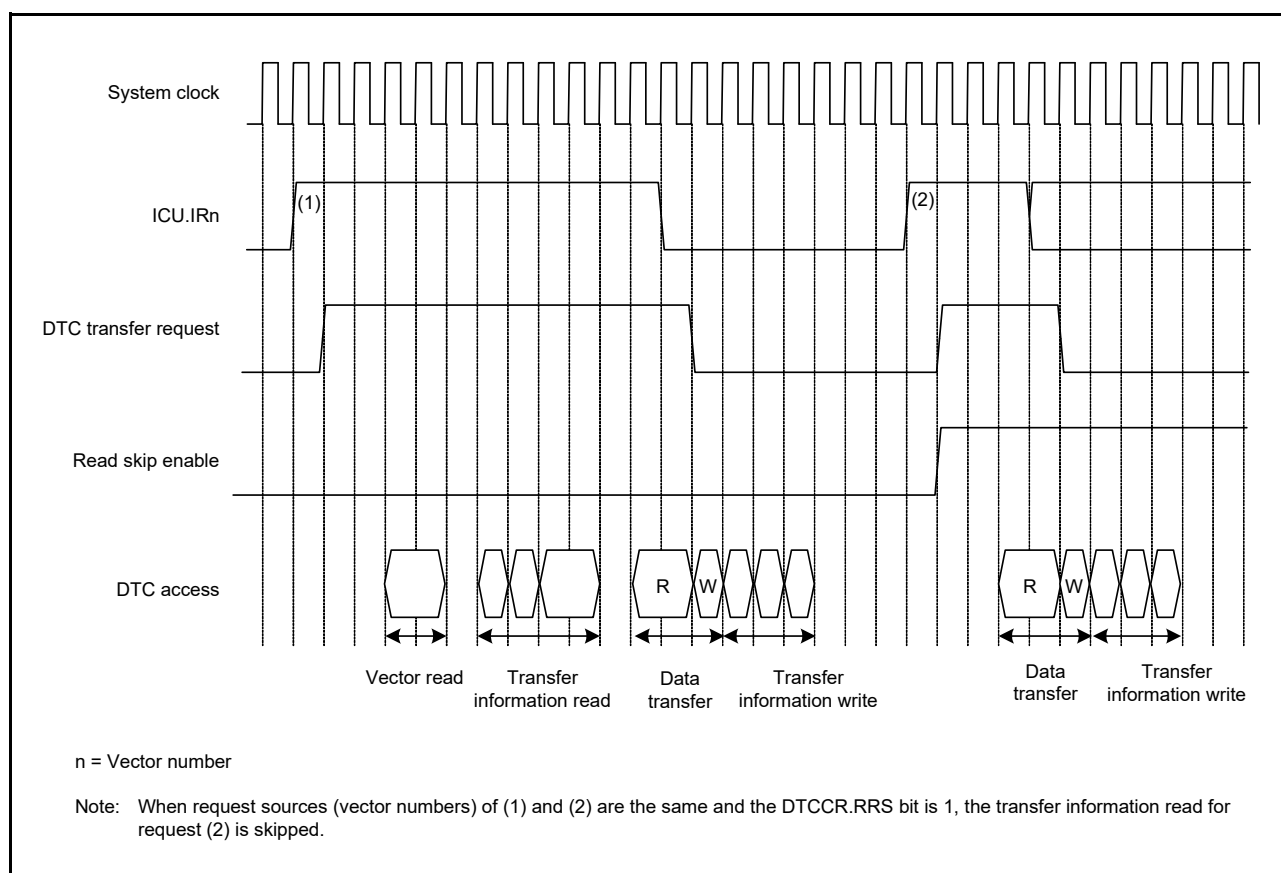


Figure 20.14 Example of Operation When Transfer Information Read Skip is Executed (Vector, Transfer Information, and Transfer Destination Data on the RAM, and Transfer Source Data on the Peripheral Module)

20.4.8 Execution Cycles of the DTC

Table 20.9 lists the execution cycles of single data transfer of the DTC.

For the order of the execution states, refer to section 20.4.7, Operation Timing.

Table 20.9 Execution Cycles of the DTC

Transfer Mode	Vector Read		Transfer Information Read			Transfer Information Write			Data Transfer		Internal Operation	
									Read	Write		
Normal	$C_v + 1$	0^{*1}	$4 \times C_i + 1^{*2}$	$3 \times C_i + 1^{*3}$	0^{*1}	$3 \times C_i^{*4}$	$2 \times C_i^{*5}$	C_i^{*6}	$C_r + 1$	C_w	2	0^{*1}
Repeat									$C_r + 1$	C_w		
Block *7									$P \times C_r$	$P \times C_w$		

Note 1. When transfer information read is skipped

Note 2. In full-address mode

Note 3. In short-address mode

Note 4. When neither SAR nor DAR is set to address-fixed

Note 5. When SAR or DAR is set to address-fixed

Note 6. When SAR and DAR are set to address-fixed

Note 7. When the block size is 2 or more. If the block size is 1, the cycle number for normal transfer is applied.

P: Block size (initial settings of CRAH and CRAL)

C_v : Cycles for access to vector transfer information storage destination

C_i : Cycles for access to transfer information storage destination address

C_r : Cycles for access to data read destination

C_w : Cycles for access to data write destination

(The unit is system clocks (ICLK) for “+ 1” in the Vector Read, Transfer Information Read, and Data Transfer Read columns and “2” in the Internal Operation column.)

(C_v , C_i , C_r , and C_w vary depending on the corresponding access destination. For the number of cycles for respective access destinations, refer to section 60, RAM, section 62, Flash Memory (FLASH), section 5, I/O Registers, and section 16.2.6, External Bus.)

20.4.9 DTC Bus Mastership Release Timing

The DTC does not release the bus mastership during transfer information read and transfer information write. While transfer information is not read or written, bus arbitration is made according to the priority determined by the bus master arbitrator.

For bus arbitration, refer to section 16, Buses.

20.4.10 Sequence Transfer

A sequence transfer can be executed on the request source that is specified in the DTCSQE register. A sequence transfer is started by setting the MRB.INDX bit to 1 and ended by setting the MRB.SQEND bit to 1. Setting the DTCOR.SQTFRL bit to 1, even during a sequence transfer, forcibly ends the transfer and the next DTC transfer request starts new sequence transfer with reference to the index table.

A sequence transfer includes the following processing.

- (1) The first data transfer is executed by referring to the DTC vector table in response to a DTC transfer request from the source specified in the DTCSQE register.
- (2) The DTC index table is referred based on the value of the lower 8 bits of the first data transferred data in (1) (sequence number).
- (3) The transfer information is read from the address obtained from the DTC index table.
- (4) A data transfer is executed in accordance with the transfer information. On completion of the data transfer, either one of the following operations is performed by using the values of bits MRB.CHNE and MRB.SQEND.
 - A chain transfer is executed when the CHNE bit is 1. The next transfer information is read. Go to (4).
 - The sequence transfer is suspended when the CHNE bit is 0 and the SQEND bit is 0. Go to (5).
 - The sequence transfer ends when the CHNE bit is 0 and the SQEND bit is 1.
- (5) When a DTC transfer request from the source specified in the DTCSQE register is accepted, the suspended sequence transfer is resumed and the next transfer information is read. Go to (4).

Note 1. When the ICU.DTCERn.DTCE bit becomes 0 based on the result of the data transfer, a DTC transfer request is not generated. Set the DTCE bit to 1 to resume sequence transfer. Refer to Figure 20.5 or section 15, Interrupt Controller (ICUD) for the conditions where the DTCE bit becomes 1.

Figure 20.15 and Figure 20.16 shows a basic sequence transfer operation.

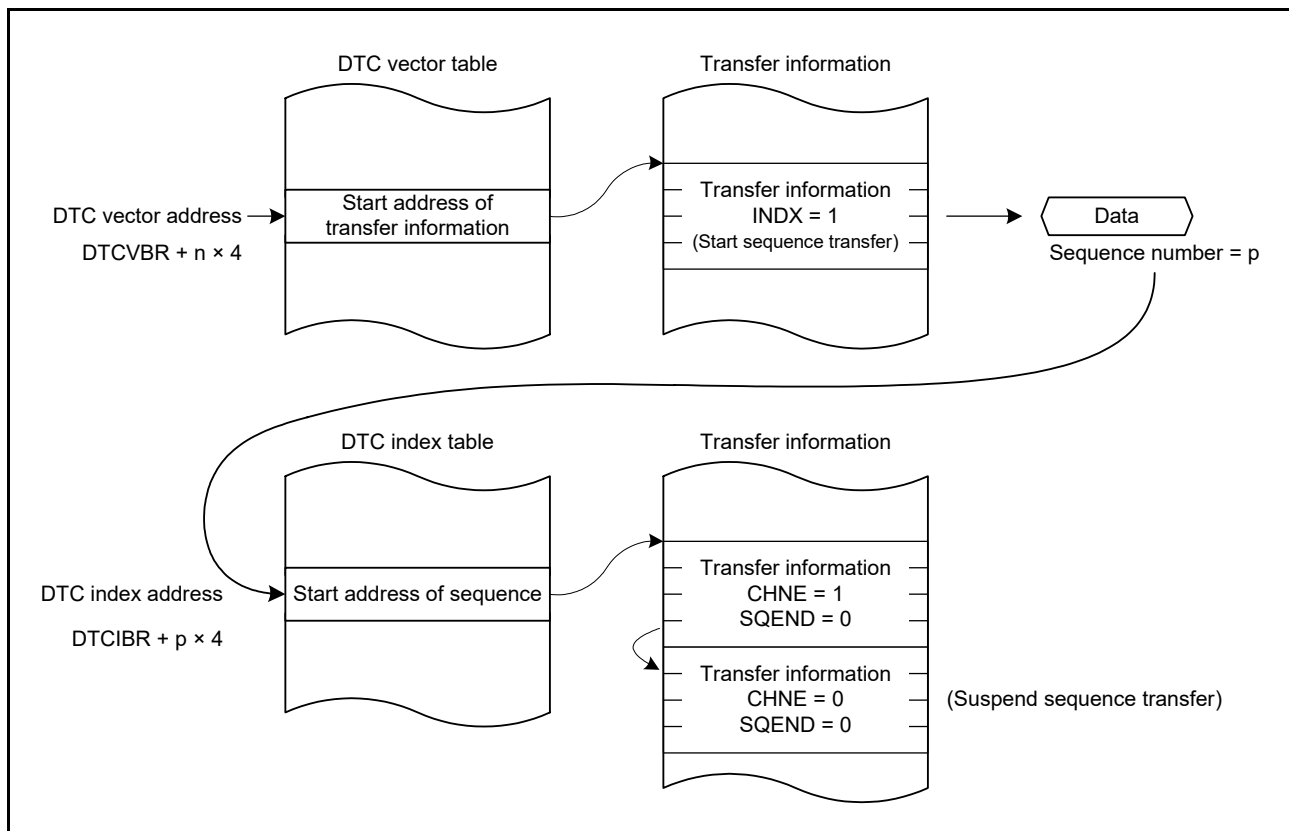


Figure 20.15 Start and Suspension of Sequence Transfer

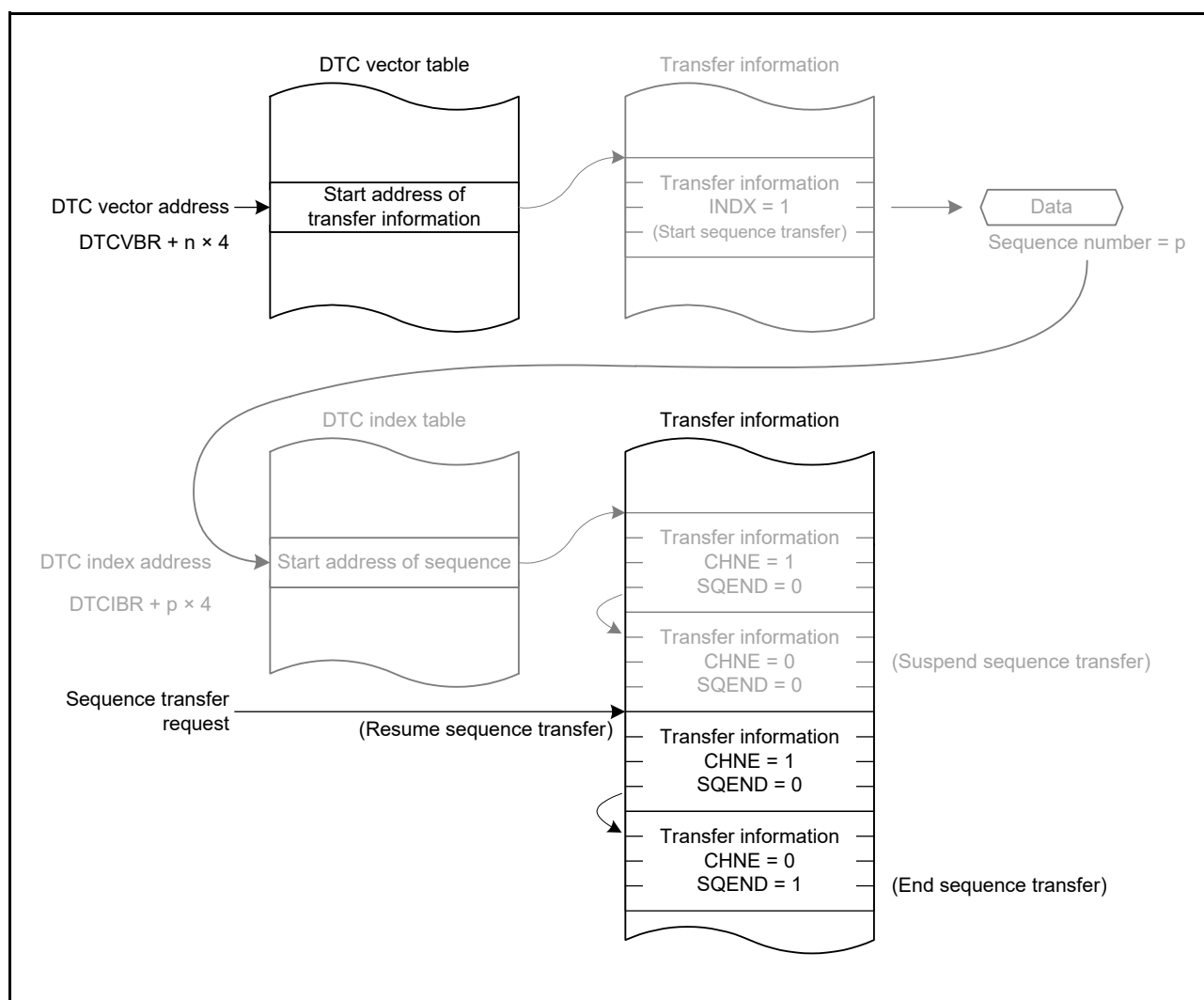


Figure 20.16 Resumption and End of Sequence Transfer

Table 20.10 lists the settings of bits CHNE, SQEND, and INDX during the sequence transfer.

Table 20.10 Sequence Transfer Process and Values of Bits CHNE, SQEND, and INDX

DTC Operations	CHNE Bit	SQEND Bit	INDX Bit
Start sequence transfer	0	0	1*1
Continue sequence transfer	1	0	0
Suspend sequence transfer*2	0	0	0
End sequence transfer	0	1	0
End current sequence transfer and Obtain new sequence number	0	1	1*1
Some other transfer (not sequence transfer)	—	0	0

Note: Do not set the values other than listed above.

Note 1. Set MRA.MD[1:0] bits to 00b (normal transfer mode) when setting the INDX bit to 1.

Note 2. When a sequence transfer is suspended, the ICU.DTCERn.DTCE bit may become 0. Set the DTCE bit to 1 to resume sequence transfer.

Even when a sequence transfer is suspended, a new sequence transfer cannot start until the suspended sequence transfer is eventually completed. When a sequence transfer request is received during suspension of the sequence transfer, the suspended sequence transfer is resumed.

20.4.11 DTC Index Table

The DTC index table is allocated to the area where its start address is configured in the DTCIBR register.

Store the start address of transfer information table p for sequence number p in the address of $\text{DTCIBR} + p \times 4$.

The upper 30 bits of the start address is set to the upper 30 bits of the DTC index. Set the CPUSEL bit to select either of reading the transfer information and starting the sequence, or output an interrupt request to the CPU without starting the sequence. For a complicated sequence that the DTC cannot handle, set the CPUSEL bit to 1 to allow the CPU to handle such a sequence.

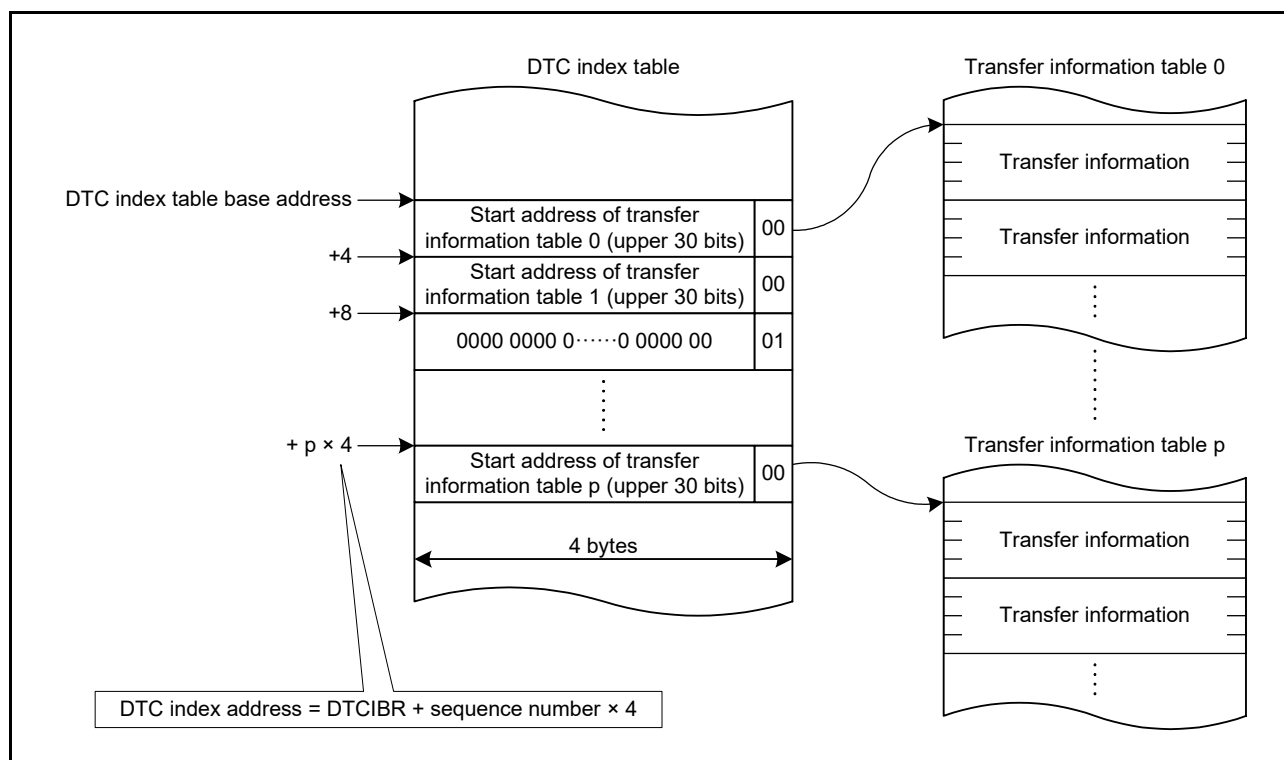
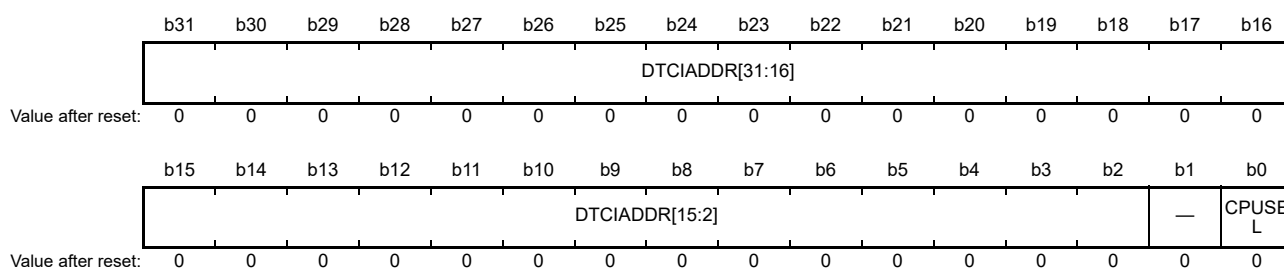


Figure 20.17 DTC Index Table

- DTC Index

Address(es): DTCIBR + p × 4



Bit	Symbol	Bit Name	Description	R/W
b0	CPUSEL	Sequence Transfer/CPU Interrupt Select	0: Continues the sequence transfer (starts the sequence) 1: Ends the sequence transfer and outputs an interrupt request to the CPU	—
b1	—	Reserved	Set this bit to 0.	—
b31 to b2	DTCIADDR[31:2]	Transfer Information Table Address	Set the upper 30 bits of the start address of the transfer information table to these bits. Writing to the upper 4 bits (b31 to b28) is ignored and the values in b31 to b28 become the same value as b27.	—

When the CPUSEL bit in the DTC index that the obtained sequence number indicates is 1, an interrupt request to the CPU is generated. At this time, the ICU.DTCERn.DTCE bit becomes 0. From this point, the interrupt request signal from the request source that is specified in the DTCSQE register is sent to the CPU, but not DTC. After completion of CPU interrupt processing, set the ICU.DTCERn.DTCE bit to 1 to enable DTC transfer request for starting the next sequence transfer.

20.4.12 Example of Sequence Transfer

Figure 20.18 shows a typical examples of a sequence transfer and Figure 20.19 to Figure 20.23 show configurations of the transfer information for the examples of the transfers in the figure.

In these examples, the interrupt source of vector number n is set as the source of the sequence transfer (DTCSQE.VECN[7:0] = n).

Once the DTC transfer request due to the interrupt source of vector number n (hereinafter referred to as “transfer request n ”) is input, the DTC refers to the DTC vector table and reads the corresponding transfer information. The lower 8 bits that have been transferred based on the transfer information become a sequence number, selecting one sequence among possible 256 sequences.

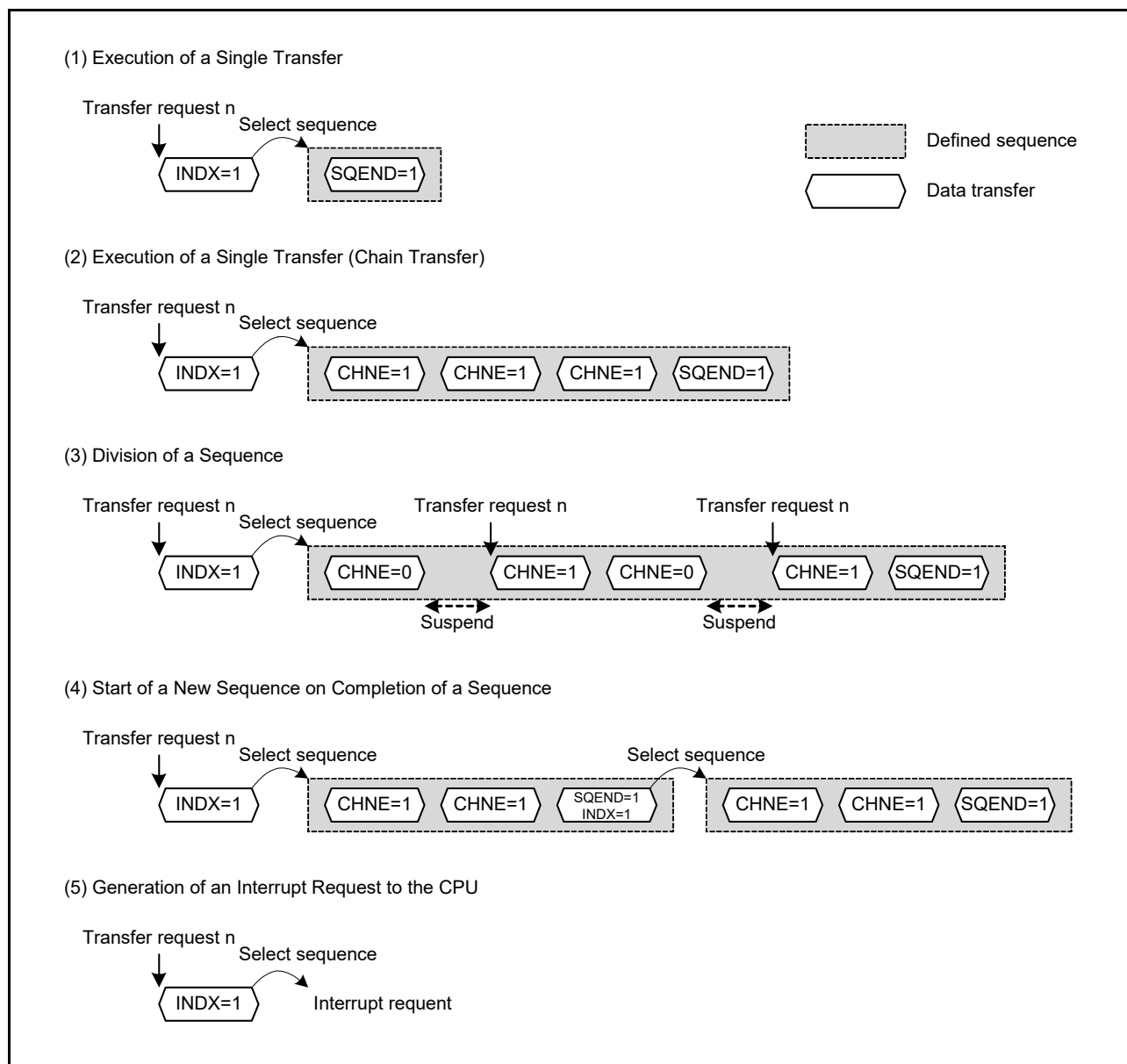


Figure 20.18 Examples of Sequence Transfers

(1) When Executing a Single Transfer

Figure 20.19 shows an example of a single transfer (normal, repeat, or block).

The DTC refers to the DTC index table, and reads the transfer information corresponding to the obtained sequence number p.

Since the values of the CHNE, INDX, and SQEND bits are 0, 0, and 1 respectively, the sequence ends after the specified transfer is executed.

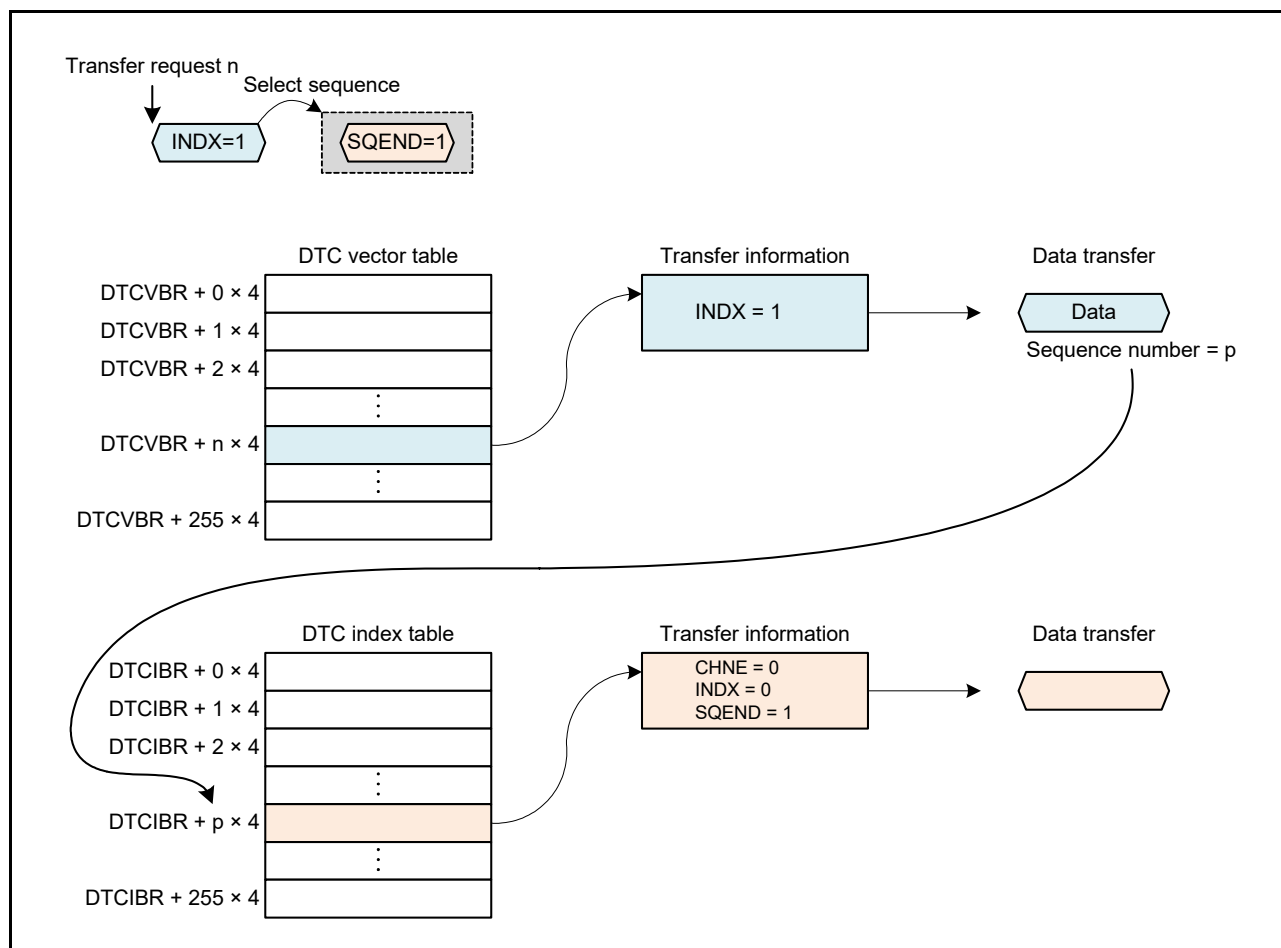


Figure 20.19 Example of a Sequence of Single Transfer

(2) When Executing a Single Chain Transfer

Figure 20.20 is an example of a sequence for a single chain transfer.

The DTC refers to the DTC index table, and reads the transfer information corresponding to the obtained sequence number q .

While the values of the CHNE, INDX, and SQEND bits are 1, 0, and 0 respectively, the specified chain transfer is executed. When the transfer information in which the CHNE, INDX, and SQEND bits are 0, 0, and 1 respectively is read, the sequence ends after the specified transfer is executed.

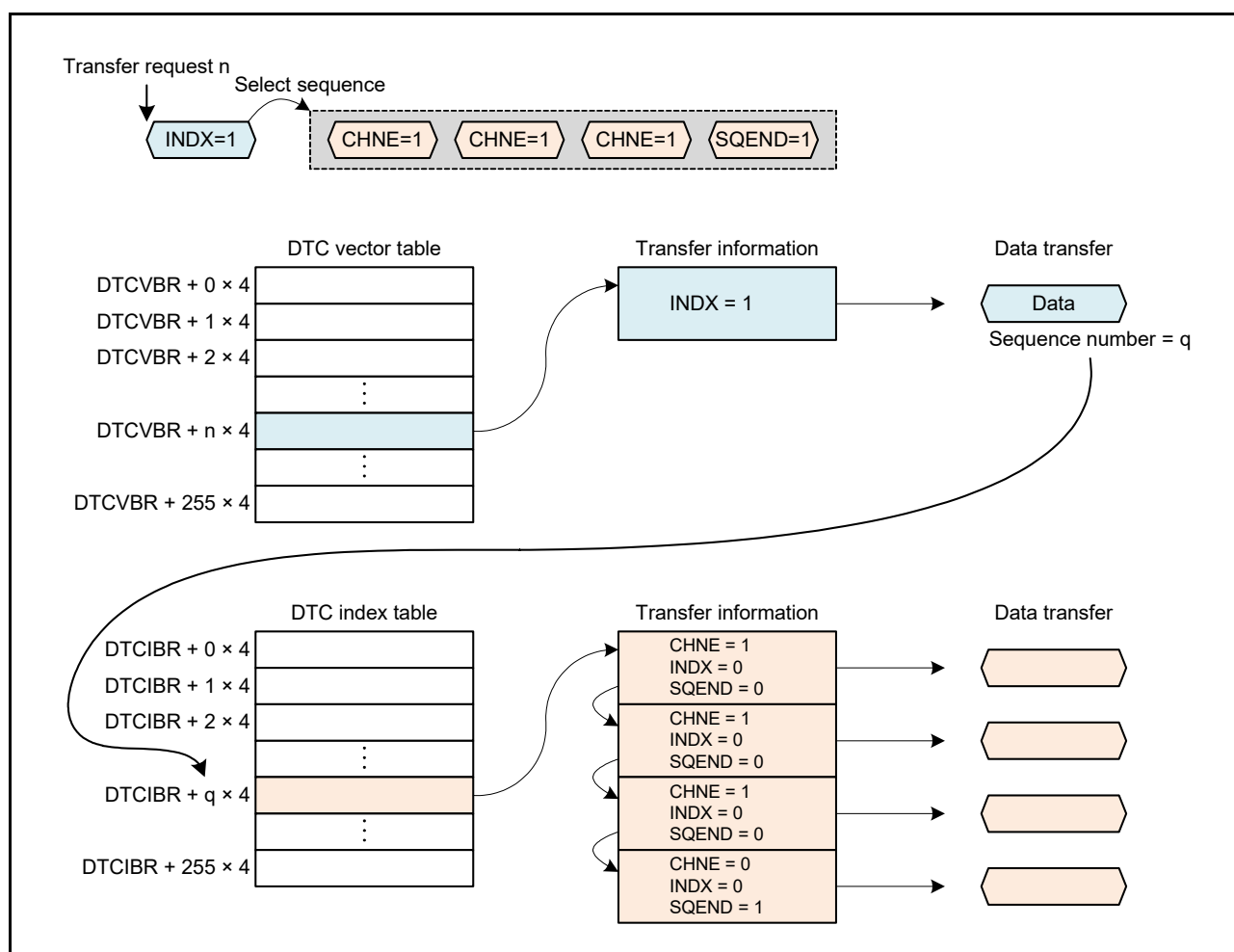


Figure 20.20 Example of Sequence of a Single Chain Transfer

(3) When Dividing a Sequence

Figure 20.21 is an example of the sequence that is divided into 3 parts.

The DTC refers to the DTC index table, and reads the transfer information corresponding to the obtained sequence number r .

Since the values of the CHNE, INDX, and SQEND bits in the transfer information are 0, 0, and 0 respectively, the sequence is suspended after the specified transfer is executed and the DTC waits for the next transfer request n .

When the transfer request n is input during a sequence transfer, the DTC vector table is not referred and the suspended sequence is resumed.

When the transfer information in which the CHNE, INDX, and SQEND bits are 0, 0, and 1 respectively is read, the sequence ends after the specified transfer is executed.

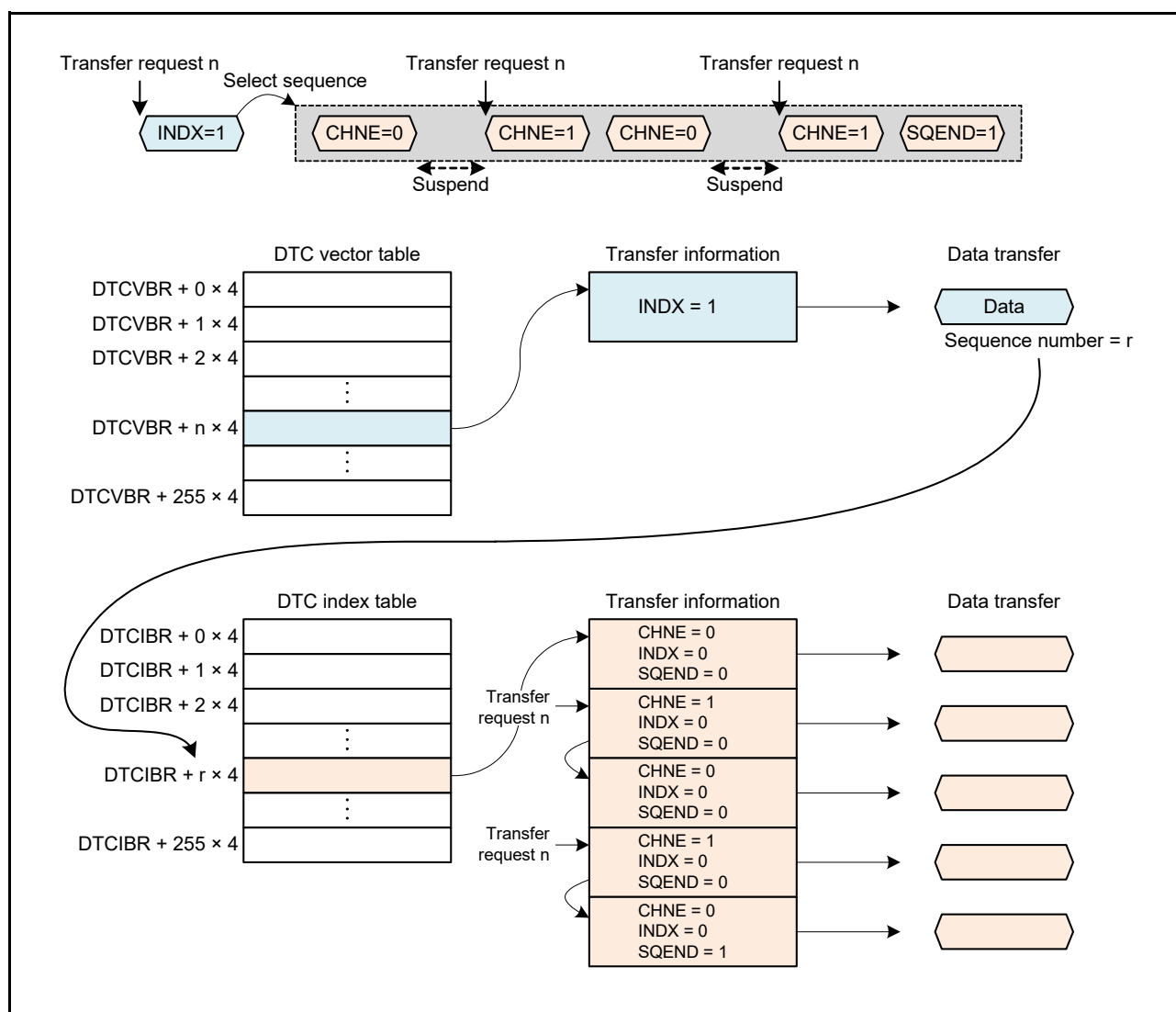


Figure 20.21 Example of Divided Sequence

(4) When Starting a New Sequence on completion of a Sequence

Figure 20.22 is an example for starting the next and new sequence on completion of the first sequence.

The DTC refers to the DTC index table, and reads the transfer information corresponding to the obtained sequence number s .

When the transfer information in which the CHNE, INDX, and SQEND bits are 0, 1, and 1 respectively is read, the specified transfer is executed, then a new sequence number is obtained from the lower 8 bits of the transferred data.

The DTC again refers to the DTC index table, and reads the transfer information corresponding to the obtained sequence number k and then starts a new sequence.

When the transfer information in which the CHNE, INDX, and SQEND bits are 0, 0, and 1 respectively is read, the sequence ends after the specified transfer is executed.

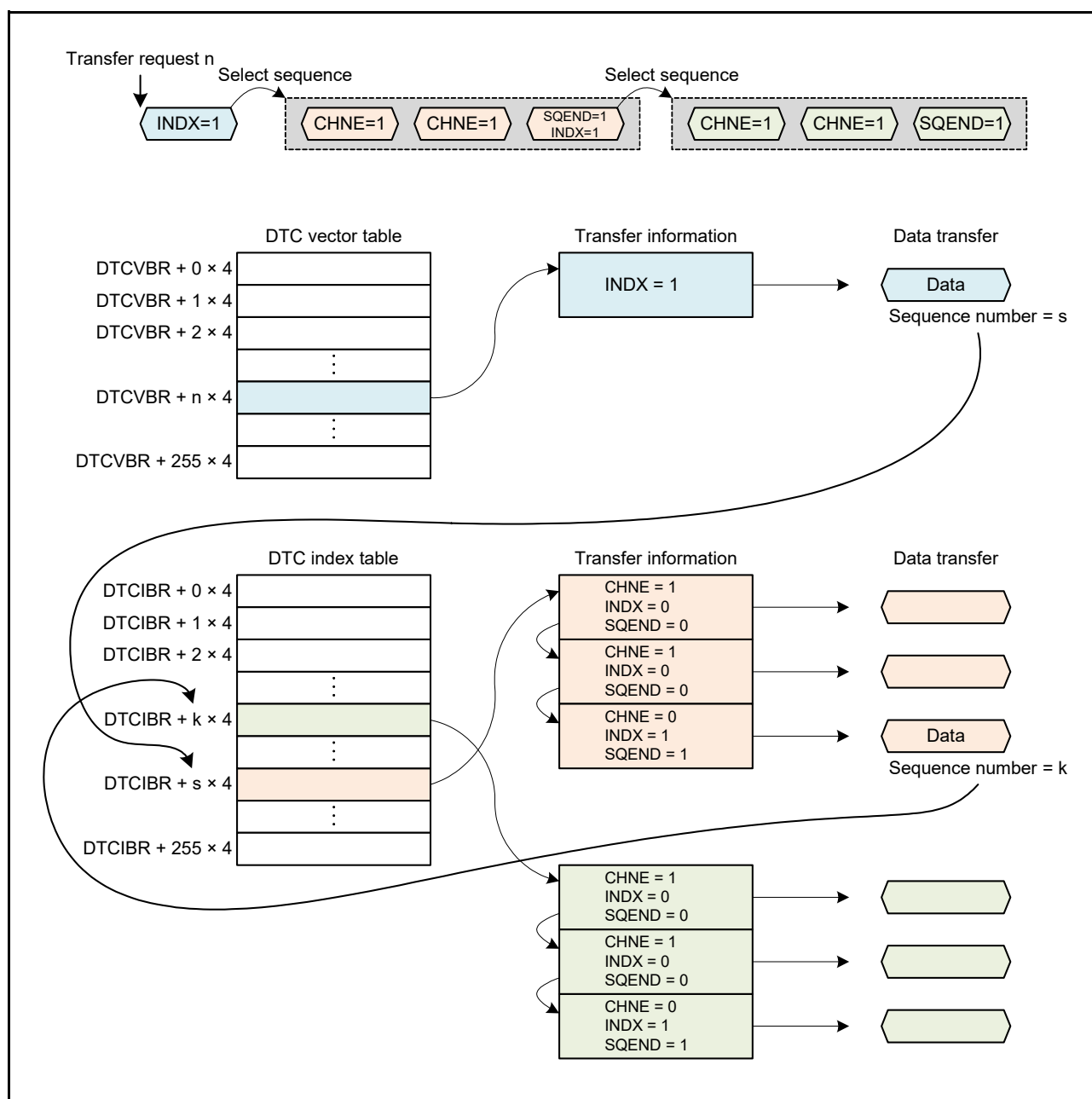


Figure 20.22 Example When Starting a New Sequence on Completion of a Sequence

(5) When Generating an Interrupt Request to the CPU

Figure 20.23 is an example of that an interrupt request is output to the CPU without starting of sequence.

The DTC obtains a DTC index that corresponds to the obtained sequence number t .

When the CPUSEL bit of the obtained DTC index is 1, the DTC ends the sequence transfer without starting the sequence, and then outputs an interrupt request to the CPU.

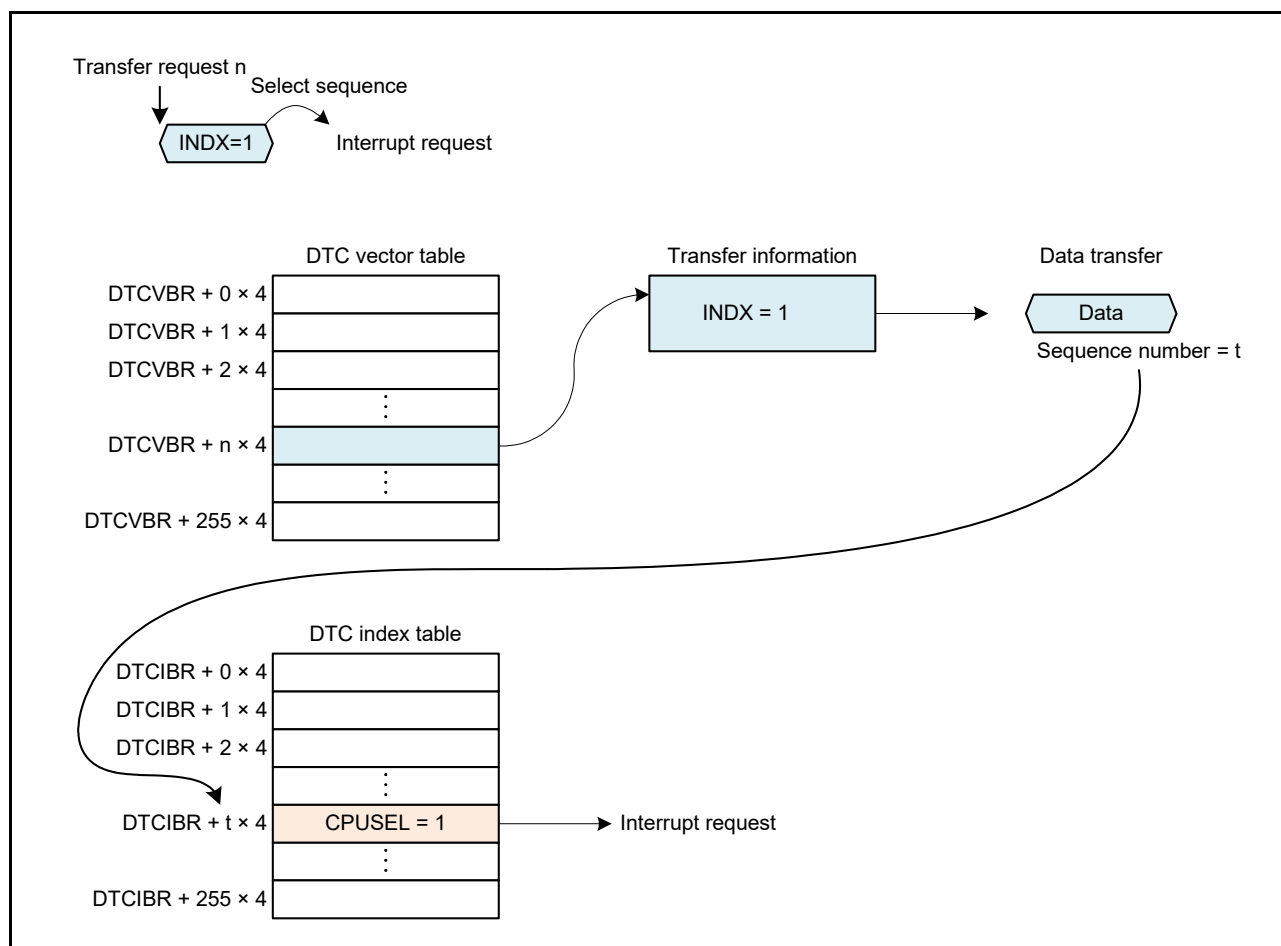


Figure 20.23 Example of Output of an Interrupt Request to the CPU

20.5 DTC Setting Procedure

Before using the DTC, set the DTC vector base register (DTCVBR). When using sequence transfer, also set the DTC index table base register (DTCIBR).

Figure 20.24 shows the procedure to set the DTC.

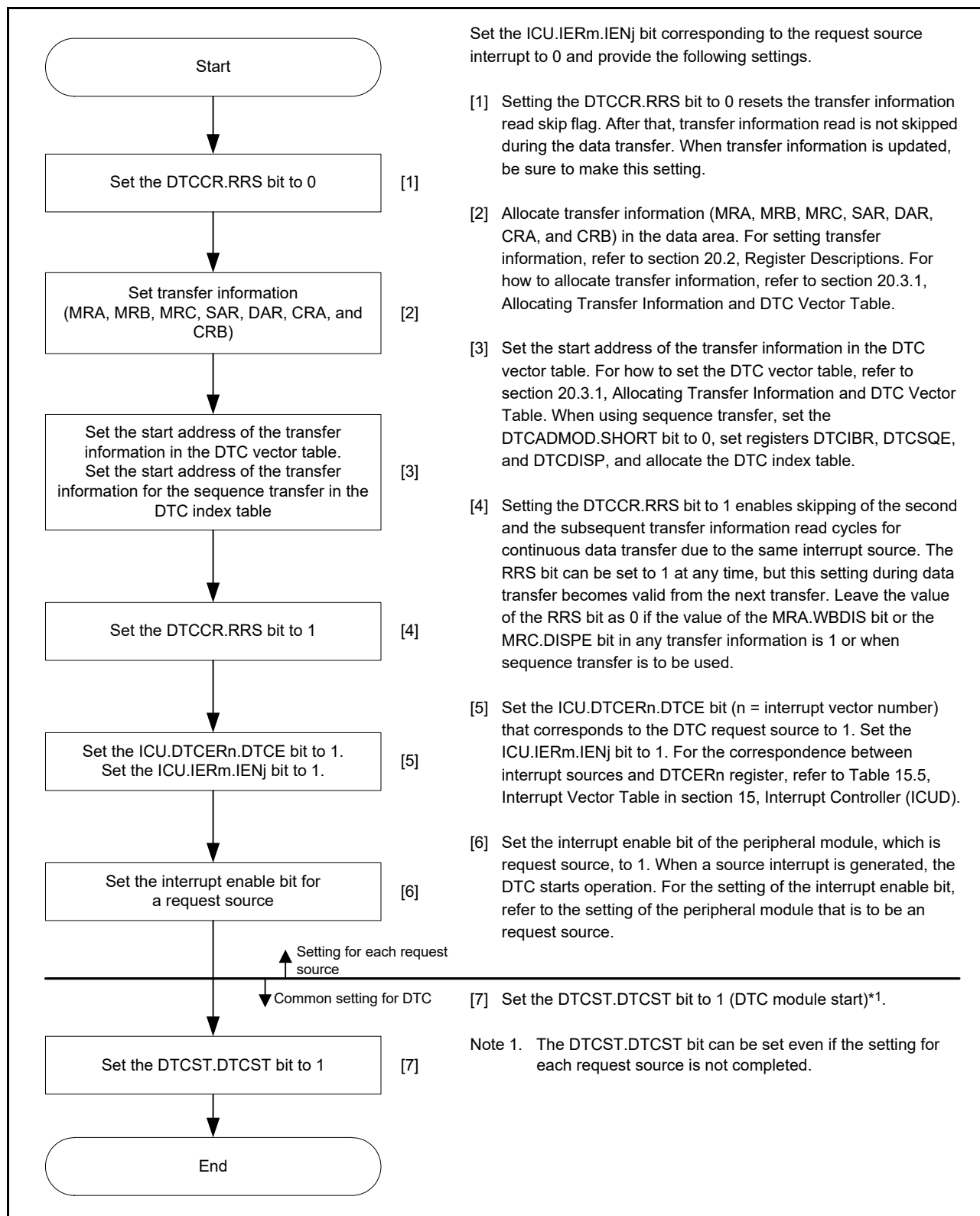


Figure 20.24 Procedure to Set the DTC

20.6 Examples of DTC Usage

20.6.1 Normal Transfer

As an example of DTC usage, its employment in the reception of 128 bytes of data by an SCI is described below.

(1) Transfer Information Setting

Set the MRA.MD[1:0] bits to 00b (normal transfer mode), the MRA.SZ[1:0] bits to 00b (byte transfer), and the MRA.SM[1:0] bits to 00b (source address is fixed). Set the MRB.CHNE bit to 0 (chain transfer is disabled), the MRB.DISEL bit to 0 (an interrupt request to the CPU is generated on completion of the specified number of data transfers), and the MRB.DM[1:0] bits to 10b (DAR is incremented after data transfer). The MRB.DTS bit can be set to any value. Set the RDR register address of the SCI in the SAR register, the start address of the RAM area for data storage in the DAR register, and 128 (0080h) in the CRA register. The CRB register can be set to any value.

(2) DTC Vector Table Setting

The start address of the transfer information for the RXI interrupt is set in the vector table for the DTC.

(3) ICU Setting and DTC Module Activation

Set the corresponding ICU.DTCERn.DTCE bit to 1 and the ICU.IERm.IENj bit to 1.
Set the DTCST.DTCST bit to 1.

(4) SCI Setting

Enable the RXI interrupt by setting the SCR.RIE bit in the SCI to 1. If a reception error occurs during the SCI receive operation, further reception is not performed. Accordingly, make settings so that the CPU can accept receive error interrupts.

(5) DTC Transfer

Every time the reception of 1 byte by the SCI is completed, an RXI interrupt is generated to start the data transfer. The DTC transfers the received byte from the RDR of the SCI to RAM, after which the DAR register is incremented and the CRA register is decremented.

(6) Interrupt Handling

After 128 times of data transfers have been completed and the value in the CRA register becomes 0, an RXI interrupt request is output to the CPU. Complete the process in the handling routine for this interrupt.

20.6.2 Chain Transfer

As an example of chain transfer by the DTC, its employment in the output of pulses by a PPG is described below.

Lower-case letters x, y, and k in this text indicate a unit, channel, and bit number, respectively.

Chain transfer is used to transfer pulse output data and change the period of the output trigger for the PPG. For the first half of the chain transfer, repeat transfer mode for transfer to the PPGx.NDRH and PPGx.NDRL registers is specified. For the second half, normal transfer mode for transfer to the MTUy.TGR registers is specified. This is because clearing of the request source and generation of an interrupt on completion of the specified number of data transfer are restricted to the second half of the chain transfer (transfer while the MRB.CHNE bit is 0).

An example of using the compare match interrupt for an MTUy.TGRA register as a request source for the DTC is provided below.

(1) First Transfer Information Setting

Settings should be made for transfer to the PPGx.NDRH and PPGx.NDRL registers. Set the MRA.MD[1:0] bits to 01b (repeat transfer mode), the MRA.SZ[1:0] bits to 01b (word transfer), and the MRA.SM[1:0] bits to 10b (SAR is incremented after data transfer). Set the MRB.CHNE bit to 1 (chain transfer is enabled), the MRB.CHNS bit to 0 (chain transfer is performed after each transfer), the MRB.DTS bit to 1 (source is repeat area), and the MRB.DM[1:0] bits to 00b (destination address is fixed). Set the SAR to the start address of the data table, the DAR register to the address of the PPGx.NDRH register, and the CRAH and CRAL registers to the size of the data table. The CRB register can be set to any value.

(2) Second Transfer Information Setting

Settings should be made for transfer to the MTUy.TGRA register. Set the MRA.MD[1:0] bits to 00b (normal transfer mode), the MRA.SZ[1:0] bits to 01b (word transfer), and the MRA.SM[1:0] bits to 10b (SAR is incremented after data transfer). Set the MRB.CHNE bit to 0 (chain transfer is disabled), the MRB.DISEL bit to 0 (an interrupt request to the CPU is generated on completion of the specified number of data transfers), and the MRB.DM[1:0] bits to 00b (destination address is fixed). The MRB.DTS bit can be set to any value. Set the SAR register to the start address of the data table, the DAR register to the address of the MTUy.TGRA register, and the CRA register to the size of the data table. The CRB register can be set to any value.

(3) Transfer Information Assignment

Place the second transfer information immediately after the first transfer information.

(4) DTC Vector Table

In the DTC vector table, set the address where the first transfer information starts.

(5) ICU Setting and DTC Module Activation

Set the ICU.DTCERn.DTCE bit corresponding to the TGIA interrupt and the ICU.IERm.IENj bit to 1.
Set the DTCST.DTCST bit to 1.

(6) MTU Setting

In the given MTUy, set the TIOR register so that the TGRA register operates as an output compare register (with output disabled) and make the TIER setting to enable TGIA interrupt requests.

(7) PPG Setting

Set the default output values in the PPGx.PODRH and PPGx.PODRL registers and the next output values in the PPGx.NDRH and PPGx.NDRL registers. Set 1 to the output bits in PORTm.PDR and PPGx.NDRH, and PPGx.NDRL. Also, select a compare match signal of the MTU as the output trigger in the PPGx.PCR register.

(8) MTU Activation

Set the MTU.TSTR.CSTk bits to 1 to start counting operation of the MTUy.TCNT counter.

(9) DTC Transfer

Every time a compare-match with the MTUy.TGRA register is generated, next output values are transferred to the PPGx.NDRH and PPGx.NDRL registers and the setting for the next output trigger period is transferred to the MTUy.TGRA register.

(10) Interrupt Handling

When the specified number of data transfers is completed (i.e. when the value of the CRA register for second transfer becomes 0), a TGIA interrupt request is issued for the CPU. Complete the process in the handling routine for this interrupt.

20.6.3 Chain Transfer When the Counter is 0

The second data transfer is performed only when the transfer counter is set to 0 in the first data transfer, and the first data transfer information is repeatedly changed in the second data transfer. Repeating this chain transfer enables transfers to be repeated more than 256 times.

The following shows an example of configuring a 128-Kbyte input buffer to addresses 20 0000h to 21 FFFFh (where the input buffer is set so that its lower address starts with 0000h). Figure 20.25 shows a chain transfer when the counter is 0.

- (1) Set normal transfer mode for input data for the first data transfer. Set the following:
Transfer source address: Fixed, the CRA register is 0000h (65,536 times), the MRB.CHNE bit is 1 (chain transfer is enabled), the MRB.CHNS bit is 1 (chain transfer is performed only when the transfer counter becomes 0), and the MRB.DISEL bit is 0 (an interrupt request to the CPU is generated on completion of the specified number of data transfers).
- (2) Prepare the upper 8 bits (in this case, 21h and 20h) of the start address at every 65,536 times of the transfer destination address for the first data transfer in another area (such as ROM).
- (3) For the second data transfer, set repeat transfer mode (source is repeat area) for rewriting the transfer destination address of the first data transfer. The transfer destination is the address where the upper 8 bits of the DAR register in the first transfer information is allocated. In this case, set the MRB.CHNE bit to 0 (chain transfer is disabled) and the MRB.DISEL bit to 0 (an interrupt request to the CPU is generated on completion of the specified number of data transfers). In this case, set the transfer counter to 2.
- (4) When a transfer request is accepted, the first data transfer is executed. When transfer is executed 65,536 times and the transfer counter of the first data transfer becomes 0, the second data transfer is started and the upper 8 bits of the transfer destination address of the first data transfer is set to 21h. At this time, the lower 16 bits of the transfer destination address and the transfer counter of the first data transfer have become 0000h.
- (5) In succession, when another transfer request is accepted, the first data transfer is executed. When transfer is executed 65,536 times and the transfer counter of the first data transfer becomes 0, the second data transfer is started and the upper 8 bits of the transfer destination address of the first data transfer is set to 20h. At this time, the lower 16 bits of the transfer destination address and the transfer counter of the first data transfer have become 0000h.
- (6) Steps (4) and (5) above are repeated infinitely. Because the second data transfer is in repeat transfer mode, no interrupt request to the CPU is generated.

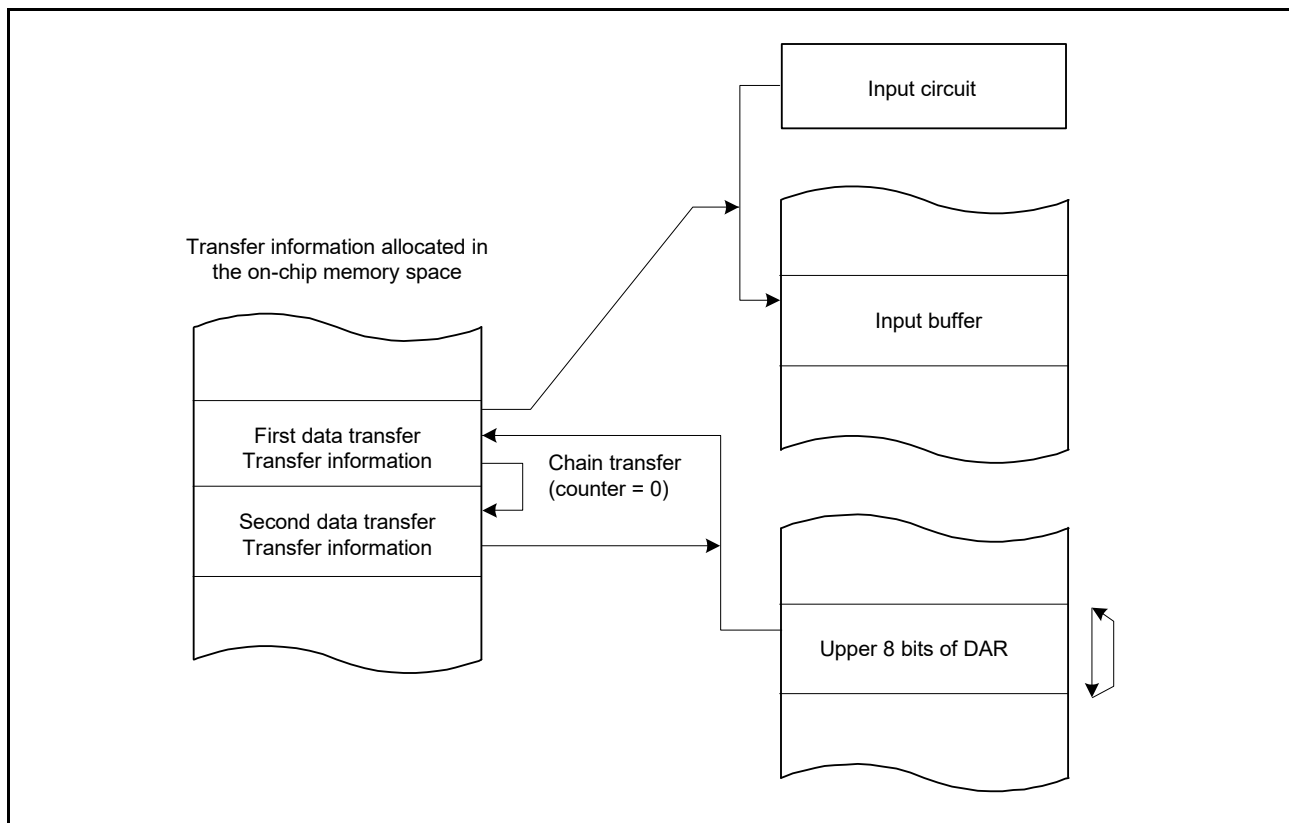


Figure 20.25 Chain Transfer When the Counter is 0

20.6.4 Sequence Transfer

The following is an example of using the SCI receive interrupt as a request source of sequence transfer.

(1) Transfer Information Settings

Set the MRA.MD[1:0] bits to 00b (normal transfer mode), the MRA.SZ[1:0] bits to 00b (byte transfer), and the MRA.SM[1:0] bits to 00b (source address is fixed). Set the MRB.CHNE bit to 0 (chain transfer is disabled), the MRB.DISEL bit to 0 (an interrupt request to the CPU is generated on completion of the specified number of data transfers), the MRB.DM[1:0] bits to 10b (DAR is incremented after data transfer), the MRB.INDX bit to 1 (start sequence transfer), and the MRB.SQEND bit to 0 (continue the sequence transfer). The MRB.DTS bit can be set to any value. Set the address of the SCIk.RDR register in the SAR register and set the start address of the RAM area which stores the data in the DAR register.

When the MRA.WBDIS bit is set to 1 (Does not write back the transfer information), the values of registers CRA and CRB are ignored.

(2) DTC Vector Table Setting

Set the start address of the transfer information for the corresponding receive data full interrupt (RXI) in the DTC vector table.

(3) DTC Index Table Setting

Set the start address of the transfer information for each sequence in the DTC index table.

(4) ICU Setting and DTC Module Activation

Set the corresponding ICU.DTCERn.DTCE bit to 1 and the ICU.IERm.IENj bit to 1. Set the DTCST.DTCST bit to 1.

(5) SCI Setting

Set the SCIk.SCR.RIE bit to 1 to enable the RXI interrupt. If a reception error occurs during the SCI receive operation, subsequent receptions are not performed. Accordingly, make settings so that the CPU can accept receive error interrupts.

(6) Start of the Sequence Transfer

On completion of reception of 1-byte data by the SCI, an RXI interrupt is generated to start the DTC. The DTC transfers the received data from the SCIk.RDR register to the RAM. The DTC looks up the DTC index table by using the value from the received data (sequence number) and continues to transfer data corresponding to the that number.

When the CPUSEL bit in the DTC index is 1, the DTC does not read the transfer information and sets the ICU.DTCERn.DTCE bit to 0. Then the DTC outputs an interrupt request to the CPU and ends the sequence transfer.

(7) During Suspension of the Sequence Transfer

Set the ICU.DTCERn.DTCE bit to 1 if the bit is 0. The DTC continues to transfer the data for every generation of the DTC transfer request in response to the corresponding RXI interrupt.

(8) End of the Sequence Transfer

Set the MRB.SQEND bit in the last transfer information of the sequence transfer to 1. After execution of this data transfer, the DTC ends the sequence transfer. The DTC starts to refer to the DTC vector table when a DTC transfer request is generated due to the next corresponding RXI interrupt.

20.7 Interrupt Source

When the DTC has finished data transfer of specified count or when data transfer with the MRB.DISEL bit set to 1 (an interrupt request to the CPU is generated each time the data transfer is performed) has been completed, an interrupt to the CPU is generated by the DTC trigger source. Such interrupts to the CPU are controlled according to the PSW.I bit (interrupt enable) of the CPU, the PSW.IPL[3:0] bits (processor interrupt priority level), and the priority level of the interrupt controller.

20.8 Event Link

The DTC outputs an event signal on completing data transfer in response to one request. When the destination for transfer is an external bus or an internal peripheral bus, however, the event signal will be output after completion of writing to the write buffer rather than after completion of writing to the actual destination for transfer.

20.9 Low Power Consumption Function

Before making a transition to the module stop state, all-module clock stop mode, software standby mode, or deep software standby mode, set the DTCST.DTCST bit to 0 (DTC module stop), and then perform the following.

(1) Module Stop Function

Writing 1 (transition to the module-stop state is made) to the MSTPCRA.MSTPA28 bit enables the module stop function of the DTC. If data transfer is in progress at the time 1 is written to the MSTPCRA.MSTPA28 bit, the transition to the module stop state proceeds after data transfer has ended. While the MSTPCRA.MSTPA28 bit is 1, accessing the DTC registers is prohibited.

Writing 0 (release from the module-stop state) to the MSTPCRA.MSTPA28 bit releases the DTC from the module stop state.

(2) All-Module Clock Stop Mode

Make settings according to the procedure under section 11.6.2.1, Transition to All-Module Clock Stop Mode, in section 11, Low Power Consumption.

If any data transfer is in progress at the time the WAIT instruction is executed, the transition to all-module clock stop mode follows the completion of the data transfer.

The DTC is released from the module stop state by writing 0 to the MSTPCRA.MSTPA28 bit following recovery from all-module clock stop mode.

(3) Software Standby and Deep Software Standby Modes

Make settings according to the procedure under section 11.6.3.1, Transition to Software Standby Mode, or section 11.6.4.1, Transition to Deep Software Standby Mode, in section 11, Low Power Consumption.

If any data transfer is in progress at the time the WAIT instruction is executed, the transition to software standby mode or deep software standby mode follows the completion of the data transfer.

(4) Notes on Low Power Consumption Function

For the WAIT instruction and the register setting procedure, refer to section 11.7.6, Timing of WAIT Instructions in section 11, Low Power Consumption.

To perform data transfer after returning from a low power consumption mode, set the DTCST.DTCST bit to 1 again.

To use a request that is generated in all-module clock stop mode or software standby mode as an interrupt request to the CPU but not as a DTC transfer request, specify the CPU as the interrupt request destination according to the description in section 15.7.3.1, Interrupt Request Destination Setting Procedure in section 15, Interrupt Controller (ICUD), and then execute the WAIT instruction.

20.10 Usage Notes

20.10.1 Start Address of Transfer Information

Set multiples of 4 for the start addresses of the transfer information to be specified in the DTC vector table. If any value other than a multiple of 4 is specified, access still proceeds with the lower 2 bits of the address regarded as 00b.

20.10.2 Allocating Transfer Information

Allocate transfer information in the memory area according to the endian of the area as shown in Figure 20.26.

For example, when writing CRA and CRB settings in 16-bit units in big endian, write the CRA setting to the address plus 8h (Ch) and the CRB setting to the address plus Ah (Eh). In little endian, write the CRB setting to the address plus 8h (Ch) and the CRA setting to the address plus Ah (Eh). When writing CRA and CRB settings in 32-bit units, allocate the CRA setting at the MSB side of the 32 bits and the CRB setting at the LSB side, and write the settings to the address plus 8h (Ch), regardless of endian.

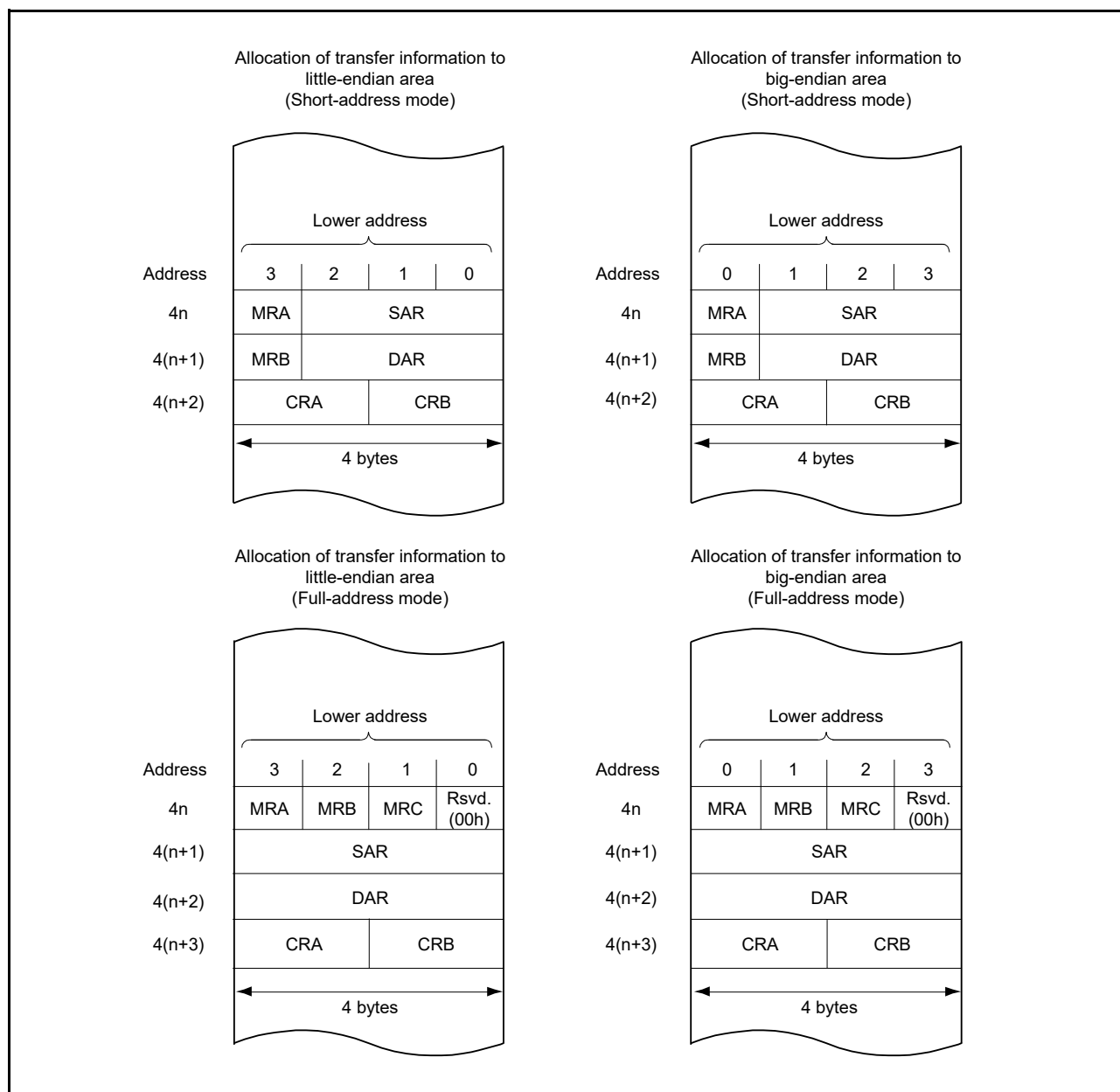


Figure 20.26 Allocation of Transfer Information

20.10.3 Setting the DTC Transfer Request Enable Register in the Interrupt Controller (ICU.DTCERn)

The DMA request should not be issued by setting the DMAC trigger select register (ICU.DMRSRm (m = DMAC channel number)) to the same vector number that has been specified by setting the ICU.DTCERn.DTCE bit to 1 (the corresponding interrupt source is selected as the DTC trigger). For details on the ICU.DTCERn and ICU.DMRSRm registers (m = DMAC channel number), refer to section 15, Interrupt Controller (ICUD).

20.10.4 Notes on Using the Sequence Transfer

When sequence transfer is to be used, make sure that the DTCADMOD.SHORT bit is 0 (full-address mode) and the DTCCR.RRS bit is also 0 (transfer information read is not skipped).

In addition, set the MRB.CHNE bit to 0 (chain transfer is disabled) when setting the MRB.INDX bit to 1 (start sequence transfer and refer the index table) or the MRB.SQEND bit to 1 (end the sequence transfer).